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A convolutional neural network based crystal plasticity finite element framework to predict localised deformation in metals

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ABSTRACT

Convolutional neural networks (CNNs) find vast applications in the field of image processing. This study utilises the CNNs in conjunction with the crystal plasticity finite element method (CPFEM). This research presents a framework that enables CNNs to make rapid and high-fidelity predictions for materials under uniaxial tension loading. The inputs to the CNN model are material hardening parameters (initial hardness and initial hardening modulus), a global tensile strain value, and microstructure with a varying number of grains, grain size, grain morphology and texture. This input selection allows performing simulations for a wide range of materials, as defined by microstructure and flow curves. The outputs of the CNN are the local stress and strain values. The proposed framework involves the following stages: feature engineering, generation of synthetic microstructures, CPFEM simulations, data extraction and preprocessing, CNN design and selection, CNN training, and validation of the trained network. The trained CNN was successfully demonstrated to predict local stress and strain evolution for the completely new dataset (test set) containing synthesised microstructures. The test set predictions were evaluated, and the median, worst, and best predictions were presented and discussed. Overall, the CNN demonstrated excellent agreement with CPFEM simulations, thus validating its accuracy. Then, the CNN was applied to predict the stress and strain evolution for AA5754 and AA6061 microstructures obtained using electron backscatter diffraction. These two microstructures were entirely new for the CNN and displayed size and grain morphology different from the synthesised microstructures. For both microstructures, the obtained stress and strain evolution predictions demonstrated excellent agreement with CPFEM simulations, thus confirming the flexibility of the trained CNN model. Then, the framework was extended to predict strain localisation and was evaluated on an AA6061 microstructure. The results presented in this research demonstrate a clear computational advantage of CNN without loss of accuracy. Finally, the research offers prospects for future advances.

1. Introduction

Metals are complex polycrystalline materials. Depending on its characterisation (texture, shape, size, and neighbours), each grain within a microstructure exhibits a distinct mechanical response. The continued advancements in material development and design

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require understanding the relationships between microstructure and flow behaviour. Microstructure constituents' effect on material performance remains an active topic of research. The developing computational simulation tools enable unravelling the impacts of material microstructure on behaviour, strength, and resistance to damage (Shamsipur et al., 2018; Nagra et al., 2017, 2020; Sabat et al., 2020; Agrawal et al., 2021), and allow for virtual design of materials with desired properties. Possible microstructures widely range in their variability due to the orientations and morphology of the grains. Therefore, there is a crucial requirement for a feasible simulation tool capable of predicting the mesoscale properties of metals subjected to a macroscopic loading.

A growing body of literature offers microstructure-based methods for modelling material behaviour. Crystal plasticity finite element method (CPFEM) based approaches are the most utilised due to their versatility in applications for predicting stress and strain partitioning, localisation, and failure for a large variety of materials. CPFEM is a physics-based method that considers the complex nature of polycrystals, such as the orientation and morphology of the grains within an aggregate. The constitutive laws, kinematics, homogenisation schemes and multiscale approaches are implemented within the CPFEM and are extended to different materials (Roters et al., 2010). The CPFEM simulation tool can help achieve accurate predictions of materials response to various loading behaviours. Examples of such applications include prediction of shear bands due to dislocation (Kuhlmann-Wilsdorf, 1999; Neale et al., 2003; Hansen et al., 2013), twinning shear (Kondo et al., 2014; Liu et al., 2021), or of non-crystallographic nature (Jia et al., 2012), or due to transformation-induced plasticity (Wong et al., 2016; Zhang et al., 2021; Connolly et al., 2022). Shear bands prediction is significant, as they indicate possible places of damage initiation. The simulations can be performed in both two- (Izadbakhsh et al., 2012; Han et al., 2014) and three-dimensions (Diard et al., 2005; Knezevic et al., 2014) for different materials, such as steel (Connolly et al., 2020; Qayyum et al., 2020), aluminium alloys (Liu et al., 2020; Iftikhar et al., 2021), magnesium alloys (Li et al., 2021; Ravaji and Joshi, 2021) and other materials (Fang et al., 2020; Zhang et al., 2020; Wroński et al., 2021). The crystal plasticity (CP) methods have proven to be a high-fidelity material modelling tool, and it continues to advance.

However, crystal plasticity models can be very computationally demanding. Different methods were implemented to expedite crystal plasticity simulations, such as: parallel computing (Inal, 2001), graphics processing unit-acceleration of CP (Mellbin et al., 2014; Eghtesad et al., 2020), using the wavelet transformation-based algorithms (Joseph et al., 2010; Chakraborty et al., 2011), and a spectral database method (Gupta et al., 2018; Alharbi and Kalidindi, 2015). A spectral database method is a data-driven approach that uses discrete Fourier transform. This method helped to improve computational time in crystal plasticity computations significantly and have been demonstrated by Kalidindi and co-workers in numerous studies (Knezevic et al., 2009; Alharbi and Kalidindi, 2015; Kalidindi et al., 2020). Kalidindi and co-workers have also presented another database approach, "Material Knowledge System", which enables the accelerated establishment of material properties (Yabansu et al., 2014; Latypov and Kalidindi, 2017; Tallman et al., 2019). New efforts that have been used to expedite CP simulations involve machine learning (ML) methods. Instead of being explicitly programmed, these methods present a function approximation tool; they capture non-linear dependencies in the datasets they are trained on (Chollet (2018)). The outstanding property of such methods is that a trained model is usually not computationally demanding, especially in comparison to numerical methods used to solve highly-nonlinear partial differential equations used in CP integration schemes. Prior to applications in CP, ML methods have been extensively explored in phenomenological modelling and computational material science. One of the earliest ML applications in material modelling was presented in 1991. Ghaboussi et al. (1991) enabled an artificial neural network (ANN), namely a multi-layer perceptron, to predict the flow curves of concrete in the state of plain stress under monotonic biaxial loading and compressive uniaxial cycle loading. The application demonstrated prominent results, and ML has been used in computational material science ever since. For example, Pidaparti and Palakal (1993) demonstrated an ANN-based material model to predict the stress-strain behaviour of composites under monotonic and cyclic loadings. Rao and Prasad (1995) used ANNs to predict flow stress of carbon steel under certain processing conditions. Many other examples of neural networks applications were found in computational material science, for example for: prediction of strengthening metal matrix composites (Mukherjee et al., 1995), prediction of flow behaviour of steel (Chun et al., 1997), design optimisation in metal forming (Kim et al., 1999), flow behaviour of particle-reinforced aluminium (Cavaliere, 2007), prediction of cold rolling texture in steel (Brahme et al., 2009). One of the notable recent publications described the use of machine learning, in particular recursive neural networks, to predict path-dependent material behaviour (Mozaffar et al., 2019). The work of Gorji et al. (2020) demonstrated the capability of feed-forward and recursive ANNs to predict stress-strain behaviour for various strain paths. ANNs also found their application in the developing field of additive manufacturing and were demonstrated to successfully predict strain partitioning during deformation in aluminium alloys manufactured using selective laser melting method (Muhammad et al., 2020).

A growing body of literature has investigated ML applications to CP models acceleration. Mangal and Holm demonstrated the power of a random forest classification algorithm to predict stress hotspots development in face-centred cubic (FCC) and hexagonal close-packed synthetic three-dimensional microstructures (Mangal and Holm, 2018b, 2019, 2018a). In their work, the authors determined whether a grain was "hot" or not depending on a number of microstructural parameters; the significance of the input features was also discussed. ANNs were applied in CP to predict flow behaviour of an aluminium alloy under a complex strain path of tension followed by simple shear (Ali et al., 2019). Yuan et al. (2018) used ANNs to predict flow behaviour and final deformation texture. The predictions could be achieved for four loading conditions (compression, tension, shear, rolling), twelve initial textures, loading rate, and Voce hardening parameters. The other applications presented: prediction of cyclic stress behaviour in steels (Miyazawa et al., 2019), relating crystal plasticity variables to the texture of titanium alloy under uniaxial tension (Acar, 2020), relating microstructure characterisation to yield stress of steel and aluminium (Tran and Wildey, 2021), prediction of texture evolution of copper under uniaxial tension (Pandey and Pokharel, 2021). In their recent work, Ibragimova et al. (2021) have utilised a feed-forward neural network method trained only on single-crystal simulations under monotonic loadings. The trained ML- and CP-based algorithm predicted the stress-strain behaviour and texture evolution for any complex strain path for any FCC crystal

orientation. The framework was successfully validated against multiple strain paths for individual grains and a polycrystal. This work was very significant, as it has shown that having a limited dataset of monotonic loadings, simulated up to 0.05 per cent of strain, can be utilised to significantly expedite CP simulations for arbitrary complex strain-paths up to any strain value. The study has demonstrated the extreme usability of ML tools for computational time savings, which is vital in modern material design.

A limited number of studies show applications of machine learning in crystal plasticity to produce finite element (FE) modelling simulation results. Convolutional neural networks (CNNs) are instrumental in making two- or three-dimensional predictions or making predictions based on two- or three-dimensional input values. In particular, [Cecen et al. \(2018\)](#) used three-dimensional convolutional neural networks to identify the relation of a microstructure of high-contrast composite material to the elastic property in the form of a component of the effective elastic stiffness tensor. [Yang et al. \(2020\)](#) used a CNN with residual connections for CP predictions. For their dataset, they performed discrete dislocation dynamics simulations for samples deformed under uniaxial tension with varying loading orientations. Their simulations were used to generate a strain map to replicate digital image correlation. Based on these strain maps, the authors were able to predict the deformation level of a sample and achieve statistical prediction of a sample's flow curve using saliency maps of hidden layers of the trained CNN. [Beniwal et al. \(2019\)](#) used CNNs to predict macroscopic stress at a strain close to a yield point based on the input image of dual-phase microstructure with varying volume fraction of phases. [Frankel et al. \(2019\)](#) utilised a hybrid of convolutional and recurrent neural networks for application in oligocrystals of an annealed austenitic stainless steel. Their work utilised the initial microstructure and the strain values as an input to the machine learning model to predict the resultant stress corresponding to the input strain, up to 6 per cent strain. In the other application for oligocrystalline materials, [Frankel et al. \(2020\)](#) utilised a similar hybrid model to predict the evolution of the dominant stress component up to 0.3 per cent strain. The input image represented the initial undeformed microstructure of synthetically generated oligocrystals. Their work demonstrated the capability of CNNs to predict overall stress partitioning based on the microstructure input up to a limited strain value. As seen in the literature, the capability of machine learning applications, particularly convolutional neural networks and their hybrid models, is yet to be investigated in their full-field predictions capabilities.

This paper proposes a convolutional network model for rapid full-field stress and strain evolution predictions for FCC polycrystalline materials. presents the methods used in this research and describes the proposed framework. The framework includes the CPFEM model for generating the simulations used for training, feature engineering, methodologies for synthetic microstructure generation, the hardening parameters sampling, and the background behind CNNs. Section 3 presents a CNN hyper-parametric study for finding an optimal architecture. Then the section validates the selected model using a test dataset with artificially synthesised microstructures unseen during training. Section 4 validates the model's flexibility for applications for real materials. The CNN was validated for AA5754 and AA6061 microstructures. Next, the CNN was assessed for the prediction of strain localisation. The section also presents the runtime comparison of CNN and CPFEM models. Finally, Section 5 summarises the work and presents the conclusions.

2. Methods & development of machine learning framework

The CNN-based framework, which enables rapid full-field stress and strain evolution prediction, is proposed in this work. The resultant CNN is trained on the CPFEM dataset and accurately predicts local stress and strain evolution for an input microstructure of a given material with defined hardening parameters. The generalised workflow of the proposed framework is displayed in [Fig. 1](#). This section includes:

- Brief description of the CPFEM model used for dataset generation,
- Methodology for artificial microstructures synthesis and sampling material hardening parameters,
- Data preprocessing methods,
- CNNs and training process background.

2.1. Crystal plasticity constitutive model and its finite element implementation

Crystal plasticity constitute models are high-fidelity models used to simulate material flow and stress and strain partitioning behaviour. The rate-dependent formulation, first introduced by [Asaro and Needleman \(1985\)](#), was used to generate CP finite element dataset for this study. In this formulation, plastic deformation occurs due to the mechanism of crystallographic slip. For an FCC metal, e.g. Aluminium, slip occurs along three unique $\langle 110 \rangle$ slip directions and in the four unique $[111]$ slip plane. Consequently, there are twelve unique slip systems for materials with FCC crystal structures. The elastic constitutive equation is expressed as a function of the Jaumann rate of Cauchy stress, $\overset{\nabla}{\sigma}$, the strain-rate tensor, D , the elastic stress tensor, $\mathcal{L}$, and the viscoplastic stress state $\dot{\sigma}^0$:

$$\overset{\nabla}{\sigma} = \mathcal{L}D - \dot{\sigma}^0 - \sigma \text{tr } D \quad (1)$$

The slip rates for each slip system, α , are found using the equation:

$$\dot{\gamma}^{(\alpha)} = \dot{\gamma}_0 \text{sgn } \tau^{(\alpha)} \left| \frac{\tau^{(\alpha)}}{g^{(\alpha)}} \right|^{1/m} \quad (2)$$

where $\dot{\gamma}_0$ is the reference shear rate, $\tau^{(\alpha)}$ is the resolved shear stress on the slip system, m is the strain-rate sensitivity index, and the parameter $g^{(\alpha)}$ is related to the slip system hardness.

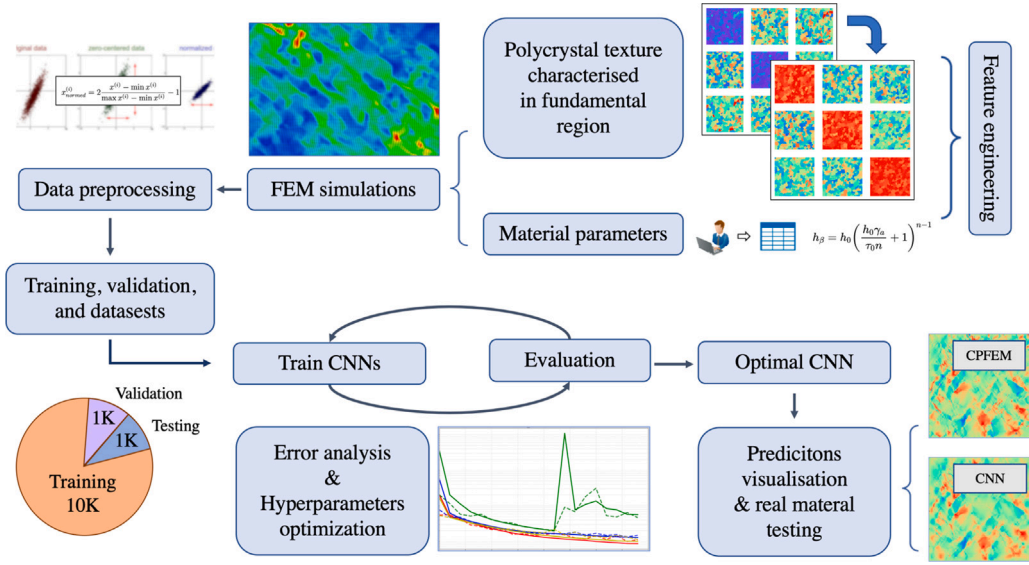


Fig. 1. Schematic representation of the proposed framework.

The single-crystal work hardening is calculated using the equation:

$$\dot{g}^{(\alpha)} = \sum_{\beta} h_{\alpha\beta} \dot{\gamma}^{(\beta)} \quad (3)$$

where $h_{\alpha\beta}$ are the hardening moduli described as:

$$h_{\alpha\beta} = q_{\alpha\beta} h_{\beta} \quad (\text{no sum on } \beta) \quad (4)$$

where $q_{\alpha\beta}$ is related to the latent hardening behaviour of a crystal, and h_{β} is the rate of a single crystal hardening. The latent hardening is expressed in a form of a matrix:

$$q_{\alpha\beta} = \begin{bmatrix} A & qA & qA & qA \\ qA & A & qA & qA \\ qA & qA & A & qA \\ qA & qA & qA & A \end{bmatrix} \quad (5)$$

where A is a 3×3 matrices fully populated with ones, and q is the ratio of latent hardening to self hardening and is set to one.

Single slip hardening used in this work is governed by a hardening law is expressed as a power-law:

$$h_{\beta} = h_0 \left(\frac{h_0 \gamma_a}{\tau_0 n} + 1 \right)^{n-1} \quad (6)$$

where h_0 is the initial hardening rate, τ_0 is the initial critical resolved shear stress (CRSS), and n is the hardening exponent. This hardening model was chosen as it is commonly used in crystal plasticity simulations.

The stress update is formulated with Eqs. (1)–(6). The texture evolution is updated using the algorithm by Raphanel et al. (2004). The initial orientation of the crystal is fed as an input to the model and then updated for each time step, in the form of a rotation matrix:

$$a(\varphi_1, \Phi, \varphi_2) = \begin{bmatrix} \cos \varphi_1 \cos \varphi_2 - \sin \varphi_1 \sin \varphi_2 \cos \Phi & \sin \varphi_1 \cos \varphi_2 + \cos \varphi_1 \sin \varphi_2 \cos \Phi & \sin \varphi_2 \sin \Phi \\ -\cos \varphi_1 \sin \varphi_2 - \sin \varphi_1 \cos \varphi_2 \cos \Phi & -\sin \varphi_1 \sin \varphi_2 + \cos \varphi_1 \cos \varphi_2 \cos \Phi & \cos \varphi_2 \sin \Phi \\ \sin \varphi_1 \sin \Phi & -\cos \varphi_1 \sin \Phi & \cos \Phi \end{bmatrix} \quad (7)$$

where $(\varphi_1, \Phi, \varphi_2)$ are the Euler angles in Bunge's notation.

The presented equations form the essence of the CP model. Their solution is very computationally demanding, especially when solved within finite element formulation. The presented CP model was incorporated into a two-dimensional FE framework (Inal et al., 2000, 2002; Brahme et al., 2011a), and its in-detail description can be found in the works by Inal et al. (2002) and Rossiter et al. (2010).

The Lagrangian formulation of the field equations was used based on the FE algorithm with the usage of convected coordinates (Inal et al., 2002). The elements of the FE mesh used in this research were akin to those presented by Wu et al. (2004). These elements have quadrilateral shapes, and each element contains four linear velocities triangular sub-element to employ a higher-order integration scheme.

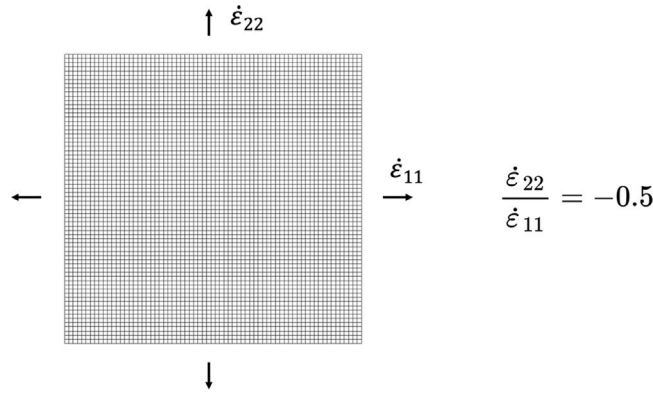


Fig. 2. Schematic representation of unit FEM cell.

The example of a sheet of aluminium alloy subjected to plane stress condition for the current application. The dimensions of the FE mesh were set to 100×100 elements. The loading ρ is imposed on the edges of FE mesh based on the following relationship:

$$\frac{\dot{\epsilon}_{22}}{\dot{\epsilon}_{11}} = \rho \quad (8)$$

where $\dot{\epsilon}_{11}$ and $\dot{\epsilon}_{22}$ are the logarithmic principal strain rates. The applied loading is similar to the uniaxial tension applied along the rolling direction and corresponds to $\rho = -0.5$. The plane stress condition was satisfied by assigning σ_{33} to zero. The computations are performed at a constant timestep $\Delta t = 0.001$ for 2500 timesteps. Fig. 2 presents a schematics of a finite element mesh undergoing proportionate loading. The presented constitutive model and its FE implementation is used within an in-house code and was employed to generate the training dataset presented in Section 2.2.

2.2. Feature engineering and dataset generation

Dataset plays an essential role in training an accurate ML model, and it is imperative that it has a sufficient amount of information representative of the set task (Zhang and Ling, 2018). To achieve the goal of predicting full-field stress and strain evolution, the minimum necessary parameters were selected as an input to the CNN model: global strain value ϵ_{11}^M , material hardening parameters, and the microstructure of the analysed material.

Due to the implementation of a constant timestep within the CPFEM model, the uniaxial macroscopic strain ϵ_{11}^M (volume-averaged) evolution is the same for all performed simulations. After 2500 steps of loading, the value of the macroscopic strain, ϵ_{11}^M , was approximately 22%, as summarised in the Table 1.

Then, the input material parameters were taken as the ones used in the hardening law (Eq. (6)): the initial hardness, τ_0 , and the initial hardening modulus, h_0 . Their values were taken in the proximity of the hardening parameters for AA5754 (Brahme et al., 2011b). The initial hardness, τ_0 , values were sampled within a range from 10 to 15 MPa, and the initial hardening modulus, h_0 , values were sampled within a range 245 to 260 MPa, as summarised in Table 1. The sampling strategy for these material parameters was implemented using Sobol sequences (Sobol', 1967; Sobol, 1976). This sampling methodology enables an optimal filling of the chosen data space. The details about the implementation of this algorithm can be found in Bratley and Fox (1988). This sampling method's main advantage is its low discrepancy property, i.e. all the sampled points in a material parameters space are sampled in an almost equidistributed fashion. In addition, this method samples the data points uniformly and quasi-randomly across the domain, which ensures that some point in the finite proximity of each combination of material parameters will be included in the training data inputs, which is beneficial for training an accurate machine learning model.

Similar to the case of hardening parameters, the microstructures, which is another input to the CNN, were selected from all possible textures in the direct-chill cast AA5754. To achieve this, the orientations within microstructures were sampled from the experimentally measured textures. The experimentally measured textures were obtained using an electron backscatter diffraction (EBSD) AA5754 map. The obtained set of individual textures contained 8750 different grain orientations. The texture data is presented in Fig. 3 displaying, (a), the extracted EBSD map, and (b) presents the pole figure of the experimentally obtained texture. Due to the symmetrical nature of crystals within FCC materials, the crystal orientation space can be decreased to the fundamental region (Kocks et al., 1998). For FCC materials, crystal orientations exhibit full octahedral symmetry, which means that each crystal orientation can have 48 different texture representations. All the orientations outside of the fundamental region zone were converted to their corresponding values within that zone. Therefore, the crystal orientations were only sampled within the fundamental region, which decreased the diversity of texture representations. The diversity of the training set's microstructures (i.e. grain size and grain morphology) was achieved using the Euclidean distance-based Voronoi tessellations algorithm. The resultant microstructures contained between 220 to 320 grains, which is an adequate number for information on microstructural and textural characteristics representing the sheet (Brahme et al., 2011b). The obtained grain morphologies were implemented into the FE mesh with the size

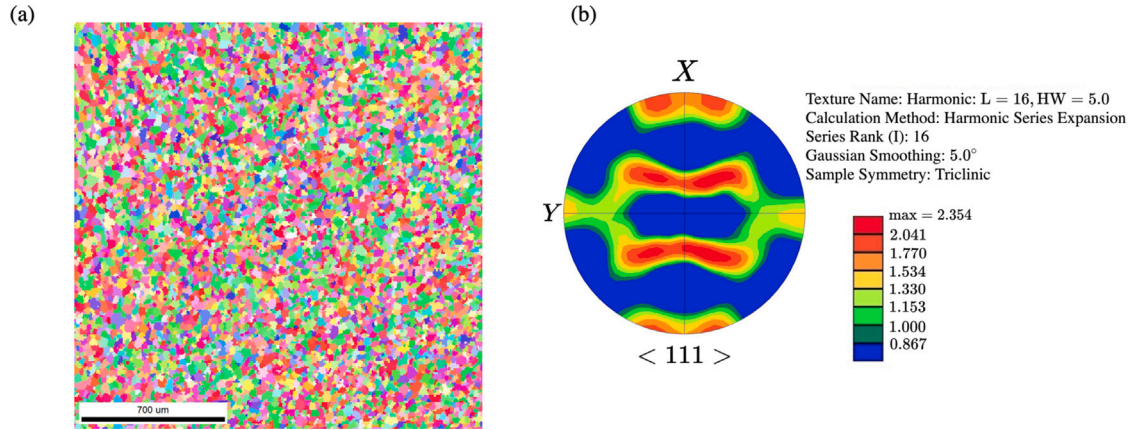


Fig. 3. The extracted data for AA5754; (a) EBSD map, (b) pole figure of the microstructure.

Table 1

Input features, their ranges, and description of the feature and the physical parameters it controls. In total, there are 12 distinct features that serve as input to the CNN model: macroscopic strain value, initial hardness, initial hardening modulus, and a microstructure.

Input variable	Range	Description
ϵ_{11}^M	0.0 ... 0.22	Macroscopic strain
τ_0	10.0 ... 15.0 MPa	Initial hardness
h_0	245.0 ... 260.0 MPa	Initial hardening modulus
a_{0j} , $i, j = 1 \dots 3$	-1.0 ... 1.0	Initial rotation matrix for texture

Table 2

Output features and their description. In total, there are 7 distinct features that serve as output to the CNN model: stress partitioning (each FE element is characterised by 3 components of stress tensor), and strain partitioning (each FE element is characterised by 4 components of strain tensor).

Predicted variable	Description
σ_{ij} , $ij = 11, 22, 12$	Stress response (MPa) (3 values due to plane stress condition)
ϵ_{ij} , $ij = 11, 22, 33, 12$	Strain response (MPa) (4 values due to plane stress condition)

of 100×100 elements. After the grain structure was meshed with N grains, N texture orientations were randomly sampled from the set of textures obtained from the AA5754 EBSD map, assigned to those grains, and stored in the form of rotation matrices. Each rotation matrix is characterised by nine components (Eq. (7)). The material microstructure was used as an input to CNN, as summarised in the Table 1.

The dataset contained 12,000 different simulations performed on unique inputs. The output of the CPFEM model were local stress and strain values (i.e. elemental values) for FE mesh, which were taken as outputs to the CNN model. Each 100th increment's data (of total 2500, as mentioned in Section 2.1) was saved for training, yielding 25 deformation timesteps saved for each simulation. The collected data was split into input and output sets. The input features included: material hardening parameters (initial hardness and initial hardening modulus), initial microstructure, macroscopic strain value ϵ_{11}^M . The output features included local stress and strain partitionings corresponding to the input macroscopic strain value ϵ_{11}^M . The input features are summarised in Table 1. The output features are summarised in Table 2. Due to the plane stress condition, the stress partitionings were presented for σ_{11} , σ_{22} , and σ_{12} stress tensor components, and the strain partitioning were presented for ϵ_{11} , ϵ_{22} , ϵ_{33} , and ϵ_{12} strain tensor components.

The distributions of stress and strain values obtained from simulations are presented by box and whisker plots. The line within a box shows the median, the ends of the box represent the lower and upper quartiles, and the end of the whiskers show the overall data range. Fig. 4 demonstrates the statistical distribution of the obtained strain values. Abscissae present the strain value ranges, and ordinates show the variation of the number of the finite elements characterised by the values in a specific range. Similarly, Fig. 5 demonstrates the statistical distribution of the obtained stress values. These observations show that data for both stresses and strains are not distributed evenly across the elements of FE meshes. The distributions differ for different variables. The CNN proposed in this research is able to capture predictions for local stress and strain evolution, and it was not required to train separate models for each output variable. The next section presents the background behind the CNN model, employed activation functions, optimisation algorithm, and regularisation methods.

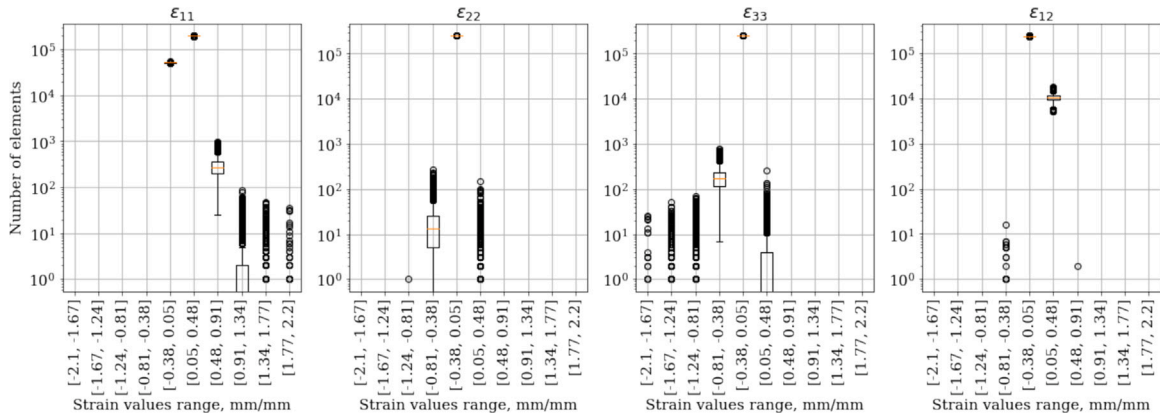


Fig. 4. Distribution of strain values in the dataset.

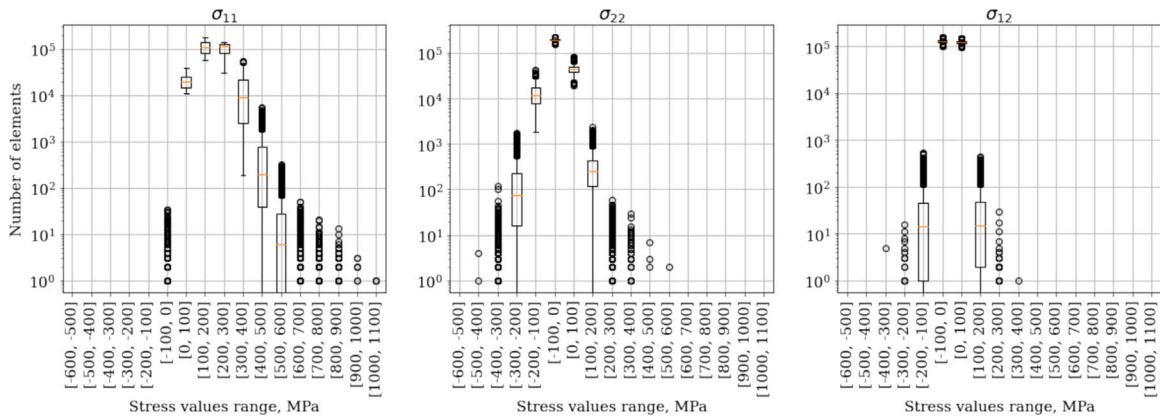


Fig. 5. Distribution of stress values in the dataset.

2.3. Convolutional neural networks

Artificial neural networks are the most common ML models. There are many different types of ANNs, such as multilayer perceptrons, convolution neural networks, and others. Previously, the capability of multilayer perceptron models to predict stress-strain evolution for single crystals, as well as the polycrystal average response, was demonstrated (Ibragimova et al., 2021). However, multilayer perceptron models would not be suitable for predicting values such as local stress and strain partitionings, as the number of trainable parameters would be too large, especially if the dimensions of a FE mesh are large.

In contrast to multilayer perceptron models, CNN models are suitable for two-dimensional input. Typically, they are applied in the field of visual imagery analysis, such as image segmentation (Milletari et al., 2016; Badrinarayanan et al., 2017) and recognition tasks (Sun et al., 2016; Khan et al., 2019). In the current work, the authors did not use image input to the CNN and instead explored the CNN's capability as a regression model to predict the local stress and strain partitionings based on the input of actual material characteristics.

In this research work, the explored CNNs consisted of: input and output layers, convolutional layers, deconvolutional layers, regularisation layers, and activation layers. The input tensor's dimension is (100, 100, 12). The first two dimensions correspond to the dimensions of the FE mesh, and the third dimension (number of channels the of input layer) corresponds to the number of input variables and includes: a channel fully populated with the macroscopic strain value ϵ_{11}^M , two channels fully populated with the material parameters τ_0 and h_0 , and 9 channels populated with corresponding components of rotation matrices. The output layer has, in total, seven channels of size 100×100 , each corresponding to the prediction stress and strain partitionings of the mesh. The predictions are made for the components σ_{11} , σ_{22} , σ_{12} , ϵ_{11} , ϵ_{22} , ϵ_{33} , and ϵ_{12} . Due to a plane-stress condition, other stresses- and strain-tensor components are equal to zero for all the elements.

Convolutional and deconvolutional layers are formed by implementing direct convolution and transposed convolution operations on square neuron layers of CNNs. The operation of direct convolution can be described with the following equation:

$$x_{ij}^{[L]} = \sum_{\xi=0}^{m-1} \sum_{\zeta=0}^{m-1} \theta_{\xi\zeta} x_{(i+\xi)(j+\zeta)}^{[L-1]}, \quad (9)$$

Table 3
Features and their normalisation methods.

Variable	Normalisation method
ε_{11}^M	Normalisation in $-1 \dots 1$
τ_0	Normalisation in $-1 \dots 1$
h_0	Normalisation in $-1 \dots 1$
a_{ij} , $i, j = 1 \dots 3$	No normalisation
σ_{ij} , $ij = 11, 22, 12$	Normalisation in $-1 \dots 1$
ε_{ij} , $ij = 11, 22, 33, 12$	Normalisation in $-1 \dots 1$

where x is the pre-nonlinearity square layer, and L is a positive integer number which denotes the number of the layer ($L \in [0, \ell]$, where $L = 0$ is the index of input, and $L = \ell$ is the index of an output layer), and θ is the filter matrix of the dimension (kernel size) $m \times m$, and it is populated with trainable parameters. Direct convolution results in the downsampling of an input layer. The transposed convolution operation is similar to direct convolution but serves as a trainable upsampling. In-detail description of convolution operations can be found in [Goodfellow et al. \(2016\)](#).

Padding parameters can be applied to control dimensions of the output of convolution operations. The term “padding” refers to adding zeros to the borders of filters prior to subjecting them to a convolution operation. This research uses two types of paddings: “same” and “valid”. “Valid” padding refers to no padding, and “same” results in dimensions of filters of input and output of convolutions being equal. In-detail explanation of the operations in the convolutional networks can be found in the work of [Dumoulin and Visin \(2016\)](#).

2.3.1. Activation function & normalisation

Activation functions are employed to add non-linear qualities to the neural networks ([Sharma, 2017](#)). In this study, two activation functions were used: the hyperbolic tangent activation function (“tanh”) and rectified linear unit (“ReLU”) activation function. The tanh activation is expressed as:

$$\tanh(x) = \frac{(e^x - e^{-x})}{(e^x + e^{-x})} \quad (10)$$

The benefit of function is that it is a continuous and differentiable zero-centred activation, with its gradient varying in different directions. This function is defined for all real numbers and ranges from -1 to 1 . The suitable normalisation for this activation function is a min–max normalisation between -1 , 1 . The following transformation achieves this normalisation:

$$x_{normed}^{(i)} = 2 \frac{x^{(i)} - \min x^{(i)}}{\max x^{(i)} - \min x^{(i)}} - 1, \quad (11)$$

where $\min x^{(i)}$ is a minimum and $\max x^{(i)}$ is a maximum value of the i th feature.

The ReLU activation function is piece-wise linear function, and is expressed as:

$$f(x) = x^+ = \max(0, x) \quad (12)$$

The additional data normalisation was not required for this activation since it was used after the batch normalisation, which serves to re-centre the weights in the network. The employed normalisation methods are shown in [Table 3](#).

2.3.2. Evaluation of artificial neural networks

The cost function, $J(\theta, b)$, and loss function, $\mathcal{L}(y_{pred}^{(i)}, y_{true}^{(i)})$, are used to evaluate the accuracy of the trained CNN. The cost function is used to compute the accuracy of the CNN. An optimisation algorithm minimises a cost function during training by correcting the weights and biases of a network ([Chollet, 2018](#)). The loss function is used to calculate the accuracy of a trained model on one sample. A cost function is calculated by averaging the sum of losses for m samples in a dataset:

$$J(\theta) = \frac{1}{m} \sum_{i=1}^m \mathcal{L}(y_{pred}^{(i)}, y_{true}^{(i)}), \quad (13)$$

Mean squared error (MSE) is used as a loss function in this research. It is expressed as:

$$\mathcal{L}_{MSE}(y_{pred}^{(i)}, y_{true}^{(i)}) = \frac{1}{n} \sum_{i=1}^n (y_{true}^{(i)} - y_{pred}^{(i)})^2, \quad (14)$$

where n is a number of elements in an output vector.

2.3.3. Optimisation algorithm

Adaptive Moment Estimation (Adam) ([Kingma and Ba, 2014](#)) is used as an optimisation algorithm in this research. It is a gradient-descent backpropagation algorithm with an adaptive learning rate. This algorithm preserves an exponentially decaying average of past gradients v_t and updates and then calculates a new value for the decaying average of past gradients:

$$\begin{aligned} m_t &= \beta_1 m_{t-1} + (1 - \beta_1) g_t, \\ v_t &= \beta_2 v_{t-1} + (1 - \beta_2) g_t^2, \end{aligned} \quad (15)$$

where m_t is the first moment of gradients, and v_t is the second moment. Then, to avoid these gradients being biased towards zero, their values are corrected according to:

$$\begin{aligned}\hat{m}_t &= \frac{m_t}{1 - \beta_1^t}, \\ \hat{v}_t &= \frac{v_t}{1 - \beta_2^t}.\end{aligned}\tag{16}$$

The network weights are then calculated as:

$$\theta_{t+1} = \theta_t - \frac{\alpha}{\sqrt{\hat{v}_t + \epsilon}} \hat{m}_t,\tag{17}$$

where β_1 , β_2 , α , and ϵ are the algorithm's hyperparameters that can be tuned during training.

2.3.4. Regularisation

One of the major objectives of training an accurate neural network is to avoid overfitting. In deep learning, the models can be very complex, and a large number of trainable parameters, while having increased capability to learn complex relations, also have the risk of overfitting on the training data. Introducing regularisation into a machine learning model can help accelerate the training process and prevent overfitting. Three different forms of the regularisation parameters were explored in the optimisation process of the neural network architecture: dropout regularisation, L_2 regularisation, and batch normalisation.

Stravastava et al. (2014) proposed the dropout regularisation method to help machine learning models prevent overfitting. The essence of the dropout method is to introduce a probability parameter that accounts for each weight's probability of being excluded from an update at each training iteration. The re-selection of trainable parameters within the regularised neural network layer is performed according to the selected probability for each iteration. This regularisation method was shown to significantly improve training and thus is included in the current study.

Another common regularisation technique that prevents overfitting in neural networks is L_2 regularisation, which is a type of weight decay technique (Krogh and Hertz, 1992; Ng, 2004). This method introduces a penalty on the cost function by adding a weighted squared magnitude of the trainable parameters to the loss function. The cost function can be presented by an equation:

$$J(\theta) = \mathcal{L}_{\text{MSE}} \left(y_{\text{pred}}^{(i)}, y_{\text{true}}^{(i)} \right) + \lambda \sum_{i=1}^{N_{\text{weights}}} \theta_i^2,\tag{18}$$

where λ is the weight parameter for L_2 regularisation.

Lastly, the efficiency of the batch normalisation method was studied in this research. Ioffe and Szegedy (2015) proposed this method to help neural networks learn faster and in a more stable fashion. The essence of batch normalisation is normalising the activation layers by standard deviation, i.e., subtracting the mean activation (re-centring) and dividing the re-centred activations by their standard deviation value. An in-depth explanation of the algorithm can be found in Bjorck et al. (2018). Batch normalisation is a popular regulariser and is often used in deep neural networks; it allows higher learning rates, makes neural networks less sensitive to the choice of trainable parameters' initial values and makes the learning process "smoother" (Santurkar et al., 2018). This method has been shown to perform well in convolutional neural networks applications (Simon et al., 2016; Ismail et al., 2019) and was included in the study conducted in this research.

3. Convolution neural network model and its validation

In this section, an optimal architecture search is performed and is presented. A series of different CNN models were first trained on a fraction of the CPFEM data and compared. The best architecture was selected based on those results. Training and test set errors were compared to assess the model's performance. Then, the best model is validated against CPFEM simulations for the completely new test set with synthetic microstructures unseen during training. The model was demonstrated to accurately predict stress and strain partitionings of FE mesh. The CNN's prediction capabilities were demonstrated for the median, worst, and best prediction cases.

3.1. Convolutional neural networks architectures

An architecture study was performed to find a suitable network capable of making accurate predictions. This section analysed and reported the effect of the number of layers, number of trainable parameters, regularisation, and choice of activations on the training and validation errors. Ten different CNN models were trained on 5000 CPFEM simulations and validated on 1000 CPFEM simulations. Using 5000 samples of 12,000 available helped achieve faster training comparison, as training deep CNNs is time-consuming. The CNNs were trained for 15 training epochs with the batch size of 4. The initial learning for the optimisation algorithm was set to 10^{-3} , and optimisation algorithm parameters were to $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$. The best model architecture was selected from all trained models based on the lowest training and validation errors. The first five of the trained CNN varied in the number of convolutional layers and filters of convolutional layers. First, the best performing network was selected out of these five trained networks. Then, this network was tested with different regularisation layers and activation functions. All the networks were compared, and the best performing network was selected for training a final model.

Table 4

Architectures of the trained models and their corresponding training and validation errors, part 1. The notation used in the architecture representation is described after each architecture description.

Architecture number	Architecture	Training/ Validation MSE
(1)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$	0.0010/0.0010
(2)	$C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32)$	0.0010/0.0013
(3)	$C_1(32) - C_2(64) - C_2^T(64) - C_1^T(32)$	0.0013/0.0014
(4)	$C_1(32) - C_2(64) - C_2(256) - C_2^T(256) - C_2^T(64) - C_1^T(32)$	0.0010/0.0025
(5)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2(256) - C_2(256) - C_2^T(256) - C_2^T(256) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$	0.0012/0.0013

Notation: $C_1(n) = \text{Conv2D}(n, \text{"same"}) \rightarrow \text{BN} \rightarrow \text{tanh}$,
 $C_1^T(n) = \text{Conv2DTranspose}(n, \text{"same"}) \rightarrow \text{BN} \rightarrow \text{tanh}$,
 $C_2(n) = (\text{Conv2D}(n, \text{"valid"}) \rightarrow \text{BN} \rightarrow \text{tanh})^2$,
 $C_2^T(n) = (\text{Conv2DTranspose}(n, \text{"valid"}) \rightarrow \text{BN} \rightarrow \text{tanh})^2$,
 where n is the number of 3×3 filters, and BN denotes batch normalisation layer.

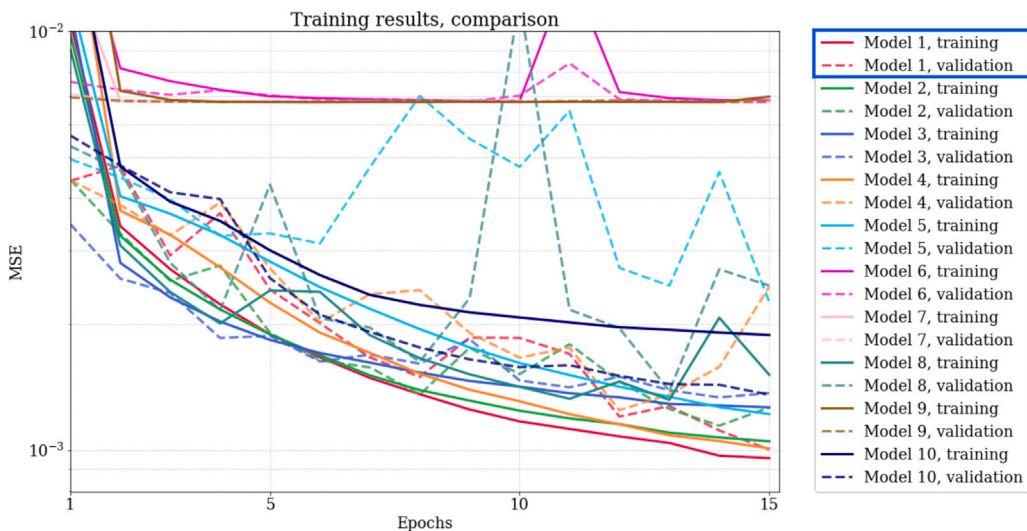


Fig. 6. Neural networks learning curves showing the evolution of losses for training and validation datasets. Each network is numerated, and each number corresponds to unique network architecture. For each ANN, the architecture and validation error are summarised in [Tables 4 and 5](#).

[Table 4](#) presents learning results for models (1) – (5). These models are of the encoder–decoder type, and they only differ by the number of constituent layers and their filters. They are composed of four different types of the architectural blocks, which are denoted as $C_1(n)$, $C_2(n)$, $C_1^T(n)$, and $C_2^T(n)$. Architecture block $C_1(n)$ is composed of a direct 2D convolutional layer with the “same” padding parameter and n filters of size 3×3 , followed by batch normalisation layer, followed by activation layer with “tanh” activation function. $C_2(n)$ is the architecture block composed of two sequential sub-blocks. Each sub-block comprises the direct 2D convolutional layer with the “valid” padding parameter and n filters of size 3×3 , followed by batch normalisation layer, followed by activation layer with “tanh” activation function. Superscript “ T ” indicates that a transpose 2D convolutional layer is used in the place of a direct convolutional layer. This notation is also explained in [Table 4](#) in a brief format. The learning curves for training and validation errors of models (1) – (5) are shown in [Fig. 6](#). The training results have shown that model (1) had the most optimal architecture out of the 5 architectures presented above. This model had 783,271 trainable parameters, which was neither the smallest number nor the largest number of parameters of all trained models. The comparison of trainable parameters for other models is following: model (2) had 762,471 trainable parameters, model (3) had 152,103 trainable parameters, model (4) had 1,795,623 trainable parameters, and model (5) had 5,358,375 trainable parameters. Therefore, it is necessary to search for suitable architecture as adding parameters and increasing the complexity of architecture do not guarantee more accurate predictive capabilities.

The second part of the architecture search studied how different activation functions and regularisation parameters affect the training results of the network. [Table 5](#) presents the training and validation errors for the networks (6)–(10) and provides descriptions for architectures. This study investigated how three different regularisation methods and two activation functions affect learning capabilities. Model (1), as the most accurate of models (1) – (5), was selected as the basis for architecture. Model (6) was trained

Table 5

Architectures of the trained models and their corresponding training and validation errors, part 2. The notation used in the architecture representation is described in the bottom of the table.

Architecture number	Architecture	Training/Validation MSE
(6)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$ Notation: $C_1(n) = \text{Conv2D}(n, \text{"same"}) \rightarrow \tanh$, $C_1^T(n) = \text{Conv2DTranspose}(n, \text{"same"}) \rightarrow \tanh$, $C_2(n) = (\text{Conv2D}(n, \text{"valid"}) \rightarrow \tanh)^2$, $C_2^T(n) = (\text{Conv2DTranspose}(n, \text{"valid"}) \rightarrow \tanh)^2$.	0.0068/0.0068
(7)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$ Notation: $C_1(n) = \text{Conv2D}(n, \text{"same"}) \rightarrow L_2(0.01) \rightarrow \tanh$, $C_1^T(n) = \text{Conv2DTranspose}(n, \text{"same"}) \rightarrow L_2(0.01) \rightarrow \tanh$, $C_2(n) = (\text{Conv2D}(n, \text{"valid"}) \rightarrow L_2(0.01) \rightarrow \tanh)^2$, $C_2^T(n) = (\text{Conv2DTranspose}(n, \text{"valid"}) \rightarrow L_2(0.01) \rightarrow \tanh)^2$.	0.0068/0.0068
(8)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$ Notation: $C_1(n) = \text{Conv2D}(n, \text{"same"}) \rightarrow \text{BN} \rightarrow \text{ReLU}$, $C_1^T(n) = \text{Conv2DTranspose}(n, \text{"same"}) \rightarrow \text{BN} \rightarrow \text{ReLU}$, $C_2(n) = (\text{Conv2D}(n, \text{"valid"}) \rightarrow \text{BN} \rightarrow \text{ReLU})^2$, $C_2^T(n) = (\text{Conv2DTranspose}(n, \text{"valid"}) \rightarrow \text{BN} \rightarrow \text{ReLU})^2$.	0.0015/0.0025
(9)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$ Notation: $C_1(n) = \text{Conv2D}(n, \text{"same"}) \rightarrow \tanh \rightarrow \text{Dropout}(0.2)$, $C_1^T(n) = \text{Conv2DTranspose}(n, \text{"same"}) \rightarrow \tanh \rightarrow \text{Dropout}(0.2)$, $C_2(n) = (\text{Conv2D}(n, \text{"valid"}) \rightarrow \tanh \rightarrow \text{Dropout}(0.2))^2$, $C_2^T(n) = (\text{Conv2DTranspose}(n, \text{"valid"}) \rightarrow \tanh \rightarrow \text{Dropout}(0.2))^2$.	0.0070/0.0068
(10)	$C_1(32) - C_2(32) - C_2(32) - C_2(64) - C_2(128) - C_2^T(128) - C_2^T(64) - C_2^T(32) - C_2^T(32) - C_1^T(32)$ Notation: $C_1(n) = \text{Conv2D}(n, \text{"same"}) \rightarrow \text{BN} \rightarrow \tanh \rightarrow \text{Dropout}(0.2)$, $C_1^T(n) = \text{Conv2DTranspose}(n, \text{"same"}) \rightarrow \text{BN} \rightarrow \tanh \rightarrow \text{Dropout}(0.2)$, $C_2(n) = (\text{Conv2D}(n, \text{"valid"}) \rightarrow \text{BN} \rightarrow \tanh \rightarrow \text{Dropout}(0.2))^2$, $C_2^T(n) = (\text{Conv2DTranspose}(n, \text{"valid"}) \rightarrow \text{BN} \rightarrow \tanh \rightarrow \text{Dropout}(0.2))^2$.	0.0019/0.0014

without batch normalisation, and the error was almost seven times larger than the error of model (1), which indicates that batch normalisation significantly improves the training process. Then, the batch normalisation regularisation method was compared with the L_2 regularisation in the model (7). The L_2 regularisation factor λ was taken equal to its standard value of 0.01. The error was comparable to one obtained for model (6) (no regularisation), which shows that L_2 does not contribute to the error decrease. The training of the model (8) realised the comparison of “tanh” and “ReLU” activation functions. In this model, batch normalisation was kept as in model (1), but all the “tanh” activation functions were replaced with “ReLU” activations. The error decay was observed, but with a lower rate when compared with the “tanh” activation function, and the resultant training error was 1.5 times higher than in the model (1). Models (9) and (10) introduced dropout regularisation without and with batch normalisation correspondingly. Using dropout without batch normalisation did not contribute to the training process, and using dropout and batch normalisation slowed down the learning process. Therefore, model (1) architecture was chosen for training on the whole dataset. Fig. 6 presents the evolution of training and validation errors during training.

To conclude, the architectures shown in Tables 4 and 5 consider various encoder–decoder types of networks. This type of network was chosen as it allows for input and output dimensions to be identical, i.e. it allows for the creation of a mapping from a finite element microstructure to finite element stress and strain partitioning. The selected networks present a broad range of parameter variation, i.e., variations in; (i) the “depth” of a CNN (number of hidden layers), (ii) “width” of a CNN (quantity of filters within hidden layers), and in (iii) activation functions and regularisation methods. An exhaustive study of different architectures would be cost prohibitive. The architecture study conducted in this research, while not being exhaustive, encompasses a range of possible architectures, and the best performing model that achieved the target validation error was employed in this manuscript. In total, ten different models were trained to find the optimal architecture. Training models (1)–(10) took approximately 250 min per model, while trained on 5000 samples. Admittedly, there were limitations to performing an extensive architecture search due to the large number of hyperparameters that can be changed and compared during training. The other limitations are the cost of collecting the data and training the models. It would be reasonable to expect more complex and deep models trained on more data to give more accurate predictions. However, the data collection and training will require more time and computational resources. The selected model has been shown to achieve excellent stress and strain evolution predictions when trained on more data, as shown in the following subsection.

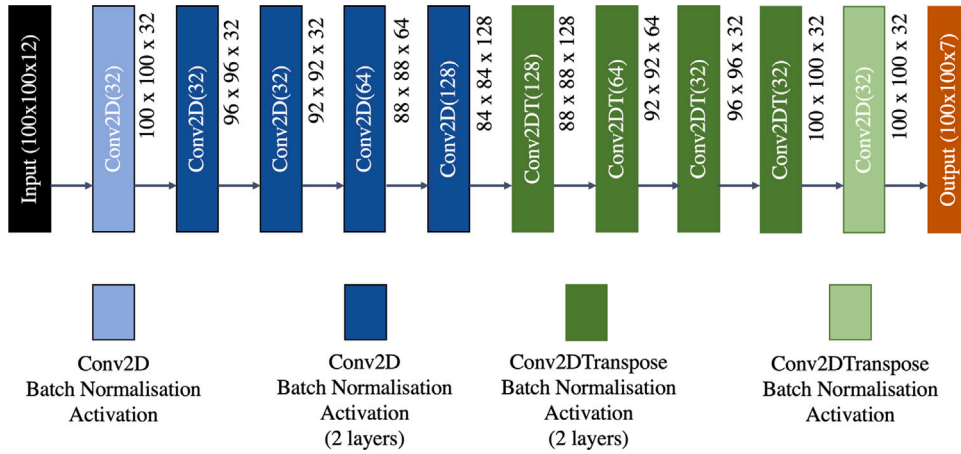


Fig. 7. The schematics of the CNN architecture. The black block shows the input to the model, and the orange block shows the model's output. The intermediate blocks of the CNN are the convolutional blocks of the CNN with implemented batch normalisation and activation layers. The dimensions next to these blocks are the resulting output dimensions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

3.2. Best model training and its evaluation on the test set

The best CNN selected in the previous subsection was used for re-training using a larger dataset. The whole dataset was composed of 12,000 samples, each corresponding to CPFEM simulation data for unique microstructure and set of material hardening parameters. The CNN was trained on 10,000 samples (training set), validated on 1000 samples (validation set), and tested on 1000 samples (test set). The validation set was used for evaluation of the CNN accuracy after each training epoch, but the information from this set was not explicitly used for training. The test set was completely unseen during the training. The network was trained for 100 epochs, and the training process took approximately 37 h. Fig. 7 shows the schematics of the architecture of the trained convolutional neural network. The black block represents the input to the CNN and shows its dimensions. Then, the input is fed to the series of encoding direct convolutional layers and then to the series of decoding transpose convolutional layers, after which the output is obtained.

The evolution of the training and validation errors is depicted in Fig. 8. The loss curves show that the training and validation errors stopped decreasing significantly at approximately the 60th epoch. At the end of the training, the training error was equal to 6.0×10^{-4} , and the validation error was equal to 5.8×10^{-4} . These errors were calculated for the normalised data and averaged across seven predicted output features, making it nearly impossible to access the network's performance for a human. To understand the predictive performance of the network, the errors for training and test sets were converted to their original scale for each individual predicted variable and assessed using the root mean square error (RMSE):

$$RMSE = \sqrt{\sum_{i=1}^N \frac{(y_i^{CP} - y_i^{CNN})^2}{N}}, \quad (19)$$

where y_i^{CP} is a CPFEM prediction for a given variable, and y_i^{CNN} its corresponding prediction made by the neural network, and $N = 10,000$ is the number of elements in each FE mesh. Two types of RMSE errors were calculated: overall RMSE averaged across all timesteps and RMSE for the final timestep of deformation.

In order to thoroughly compare the predictions of the CNN and the CPFEM models, the equivalent von Mises stress and strain values were calculated and compared for CNN and CPFEM predictions. The errors for equivalent von Mises stress and strain were obtained by taking the difference between the equivalent von Mises values of the two models. The von Mises strain error can be expressed as $\epsilon_{error}^{VM} = \epsilon_{CPFEM}^{VM} - \epsilon_{CNN}^{VM}$, where ϵ_{CPFEM}^{VM} is the von Mises strain obtained for CPFEM predictions, and ϵ_{CNN}^{VM} is the von Mises strain for CNN predictions. The von Mises stress error is calculated similarly. The equivalent stress and strains were assessed for the training and test set sample. Fig. 9 displays that the equivalent stress error values are primarily situated in the range $[-10, 10]$ MPa, with the median error being close to zero MPa. The error values for the equivalent stress confirm that the model is accurate for the overall stress predictions. The equivalent strain errors mainly were situated within the $[0, 0.2]$ range of strain, with median values close to zero (under 0.01). These results show that the trained neural network model accurately predicted tensorial strain components.

Further, the median, highest (the “worst” case), and lowest (the “best” case) error cases were visualised for the test set predictions. Fig. 10 presents the CPFEM simulation and CNN prediction results comparison for the median case. The input microstructure is shown on the top left of the figure and is coloured using inverse pole figure (IPF) colouring. The bottom left of the figure presents the stress–strain response averaged across all elements in the FEM mesh. The averaged stress–strain response RMSE is 1.93 MPa for the median case, demonstrating the trained network's excellent predictive capabilities. To the right of the input

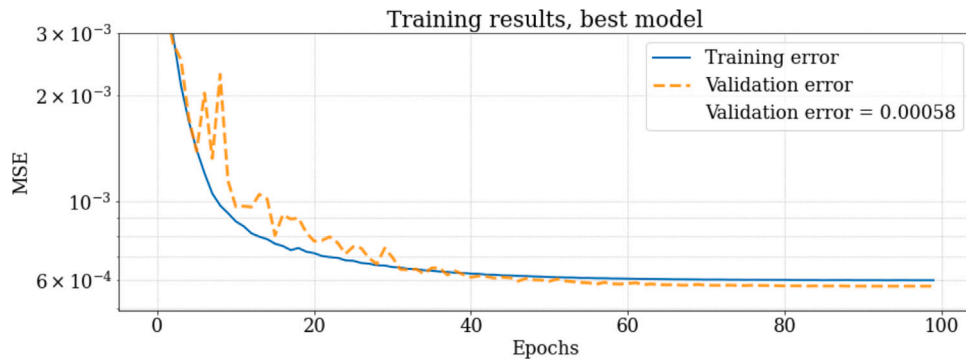


Fig. 8. The evolution of the training and validation mean squared errors for the selected CNN model.

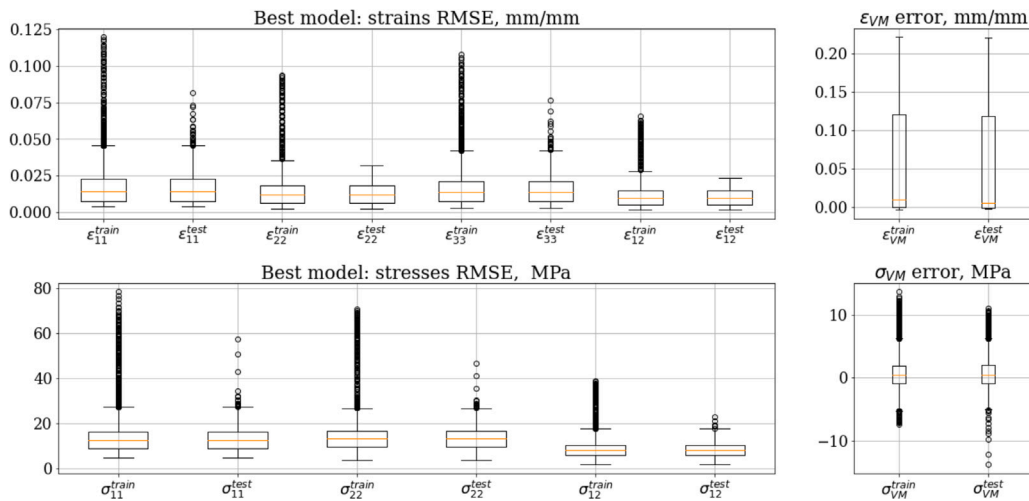


Fig. 9. Best trained model results: root mean squared errors for all variables predictions (left side of the figure) and von Mises difference error for equivalent stress and strain (right side of the figure).

texture and stress–strain curve, the figure displays the evolution of the ϵ_{11} strain component partitioning. The top row shows the CPFEM predictions compared to the CNN predictions in the bottom row. Strain partitioning of FEM mesh is compared for 6%, 12%, 17%, and 22% of strain. CP and CNN predictions results look nearly identical when compared to each other, for all deformation steps, except for minor deviations. The RMSE errors for all predicted variables are shown in the Tables 6 and 7. Table 6 reports the RMSE errors calculated over the whole history of the deformation, and Table 7 calculated for the last increment of the deformation. The RMSE error for σ_{11} stress component was equal to 16.14 MPa when calculated for all timesteps, which is equal to approximately 5% of the maximum averaged stress, and it was equal to 24.32 MPa for the last timestep which is equal to approximately 8% of the maximum averaged stress. The corresponding errors for ϵ_{11} were equal to 0.02 and 0.03, which confirm the excellent agreement of CPFEM and CNN results. The results for the median error show that the overall performance of the trained network is very accurate.

Secondly, the test set prediction with the highest error was analysed. Similarly to the median case, Fig. 11 shows the input texture, average stress–strain response, and strain-partitioning evolution for ϵ_{11} component. The input microstructure deviates from most of the textures in the training set, such that the strain localisation occurs during the deformation. This localisation results in stress unloading, as observed in the average stress–strain response. The average stress–strain response RMSE had a value of 11.55 MPa. This error is higher than the median case error, as the macroscopic stress does not capture the global softening predicted by crystal plasticity. Even though the network was not able to predict the exact intensity of ϵ_{11} strains in the localisation region, the CNN was able to identify its location accurately, as identified in Fig. 11. The rest of the strain partitioning is predicted accurately and is in excellent agreement with CPFEM results. For σ_{11} , The RMSE error was equal 22.69 MPa for the values across all timesteps of the deformation (see Table 6), and the RMSE for the last timestep was equal to 57.29 MPa (see Table 7). The RMSEs for other variables are also reported in the Tables 6 and 7. Even though this test example is a test set outlier in terms of stress–strain behaviour, the CNN predictions were very accurate up to the point of localisation occurrence. Due to such outliers in the training and test sets, several higher errors can be observed in Fig. 9 in training and test set errors for all variables.

Finally, the test case prediction with the lowest error (i.e. most accurate prediction) was visually assessed, and the RMSE for all the variables were evaluated for this case. Similarly to previous cases, Fig. 12 displays the initial microstructure in the IPF colouring

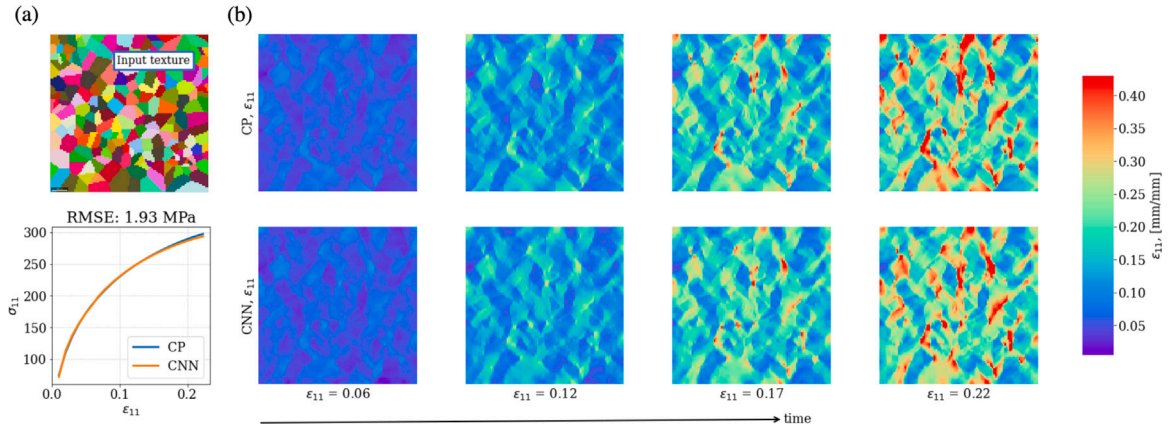


Fig. 10. Test set prediction with the median error; (a): input microstructure and flow-curve for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh; (b): ϵ_{11} strain partitioning evolution for four strain levels. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

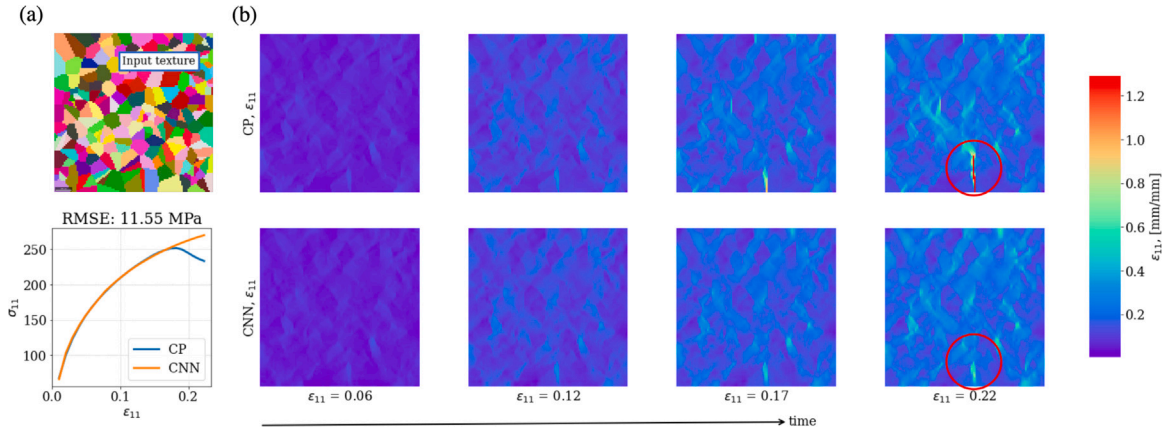


Fig. 11. Test set prediction with the highest error; (a): input microstructure and flow-curve for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh; (b): ϵ_{11} strain partitioning evolution for four strain levels.

Table 6

Element-wise RMSE errors calculated across all timesteps for all predicted variables.

Case	σ_{11} , MPa	σ_{22} , MPa	σ_{12} , MPa	ϵ_{11}	ϵ_{22}	ϵ_{33}	ϵ_{12}
Median	16.14	16.14	10.23	0.02	0.02	0.02	0.01
Worst	22.69	20.39	11.42	0.04	0.03	0.03	0.01
Best	15.37	15.61	10.28	0.02	0.01	0.02	0.01

scheme. Average stress predictions obtained with CNN show exceptional agreement with the averaged crystal plasticity response. The average stress response's RMSE error over the whole deformation was equal to 1.35 MPa. The strain partitioning evolution predicted by CNN is in excellent agreement with CPFEM simulation results. The neural network model predicted all the strain hot spots with the exception of slight intensity underprediction that can be observed in the final timestep of the deformation. The RMSE errors for σ_{11} stress component across all timesteps was equal to 15.37 MPa (see Table 6) which is comparable to the median case. For the last timestep the σ_{11} RMSE between CNN and CPFEM predictions was equal to 23.58 MPa (see Table 7) which is also comparable to the median case error. The errors for ϵ_{11} are identical for best and median cases (Tables 6 and 7) confirm excellent predictive capabilities of the CNN model.

To conclude, the results for the test set have demonstrated a good accuracy of the trained neural network. The test set's microstructures were synthetically generated using the same methodology as the training set microstructures. The test set was entirely new for the CNN model, and the neural network demonstrated excellent accuracy. The following section validates the flexibility of the trained CNN by using actual material microstructures.

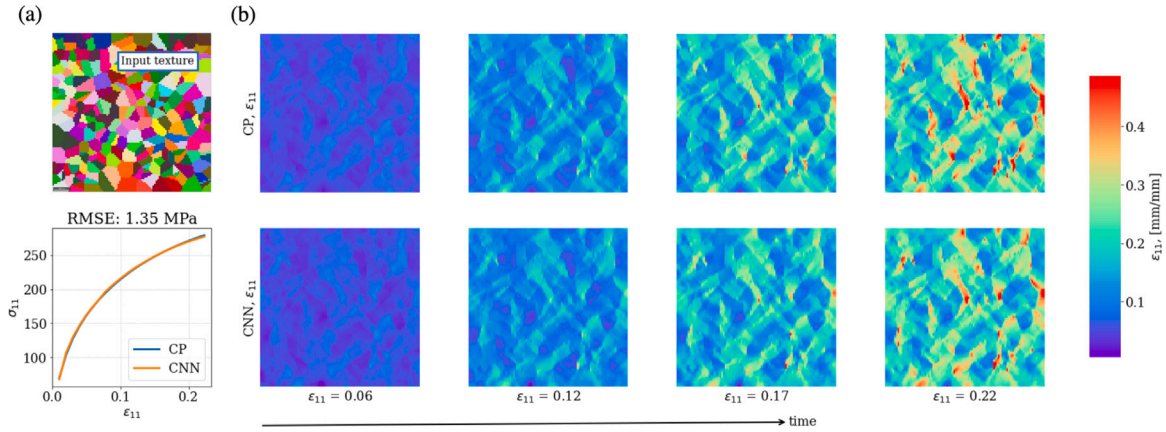


Fig. 12. Test set prediction with the lowest error; (a): input microstructure and flow-curve for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh; (b): ϵ_{11} strain partitioning evolution for four strain levels. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 7

Element-wise RMSE errors calculated for the last timesteps for all predicted variables.

Case	σ_{11} , MPa	σ_{22} , MPa	σ_{12} , MPa	ϵ_{11}	ϵ_{22}	ϵ_{33}	ϵ_{12}
Median	24.32	22.33	15.05	0.03	0.03	0.03	0.02
Worst	57.29	46.46	22.85	0.08	0.03	0.08	0.02
Best	23.58	21.71	14.82	0.03	0.03	0.03	0.02

4. Application for real materials

This section validates the CNN against CPFEM prediction for two completely new actual material microstructures: AA5754 and AA6061. The test case of AA5754 microstructure is described in detail in [Brahme et al. \(2011a\)](#), and the test case for AA6061 microstructure is described in detail in [Li et al. \(2022\)](#). For brevity, a high level of details about these materials was left out of this paper, as these details are discussed in [Brahme et al. \(2011a\)](#) and [Li et al. \(2022\)](#). Then, the extension of the model to strain localisation and stress softening response prediction was presented. Finally, the runtime comparison of CNN and CPFEM models was performed and limitations of the model were discussed.

4.1. Application to AA5754 and AA6061

This subsection compares the CPFEM and CNN predictions for the microstructures of AA5754 and AA6061. These results show the full predictive capability of the proposed framework and prove that training the CNN on the synthesised microstructure data can help achieve accurate machine learning predictions for the microstructures of real materials. For each material, the input microstructure is presented, and the predictions are visualised for ϵ_{11} strain and σ_{11} stress components. The strain and stress partitionings evolution were compared for CPFEM simulation result and CNN prediction for 6%, 12%, 17%, and 22% macroscopic strain, ϵ_{11}^M , values. Stress and strain responses averaged across all grains are presented. Two error metrics are used to estimate the accuracy of the predictions. The first error metric is an element-wise difference between CPFEM and CNN predictions, and it was visualised using the histogram method. This difference error (DE) is calculated using the equation:

$$DE = y^{CP} - y^{CNN}, \quad (20)$$

where y^{CP} is the CPFEM prediction and y^{CNN} is the prediction by the neural network. Then, an error map is employed to visualise the error between CPFEM and CNN predictions. The error map displays the absolute value of the difference error in the form of a heat map for ϵ_{11} strain and σ_{11} stress values. The white colour in error maps represents no error, and the black represents the maximum error of a given prediction. The absolute difference error (ADE) was calculated using the following equation:

$$ADE = |y^{CP} - y^{CNN}|, \quad (21)$$

where y^{CP} is the CPFEM prediction and y^{CNN} is the prediction by the CNN. Finally, the RMSE error metric was employed to calculate the error across all timesteps and for the last timestep. The RMSE were calculated for all tensorial components of strain and stress and are presented in the [Tables 8 and 9](#). The CPFEM and CNN predictions for σ_{22} , σ_{12} , ϵ_{22} , ϵ_{33} , and ϵ_{12} were compared visually for the last timestep, and the RMSE errors for all timesteps and for the last timesteps are reported in the [Tables 8 and 9](#).

First, the AA5754 microstructure was used to test the flexibility of the predictive capabilities of the CNN. [Fig. 13](#) shows: (a) the initial microstructure of the material, comparison of the averaged ϵ_{11} strain, histogram of element-wise difference errors at

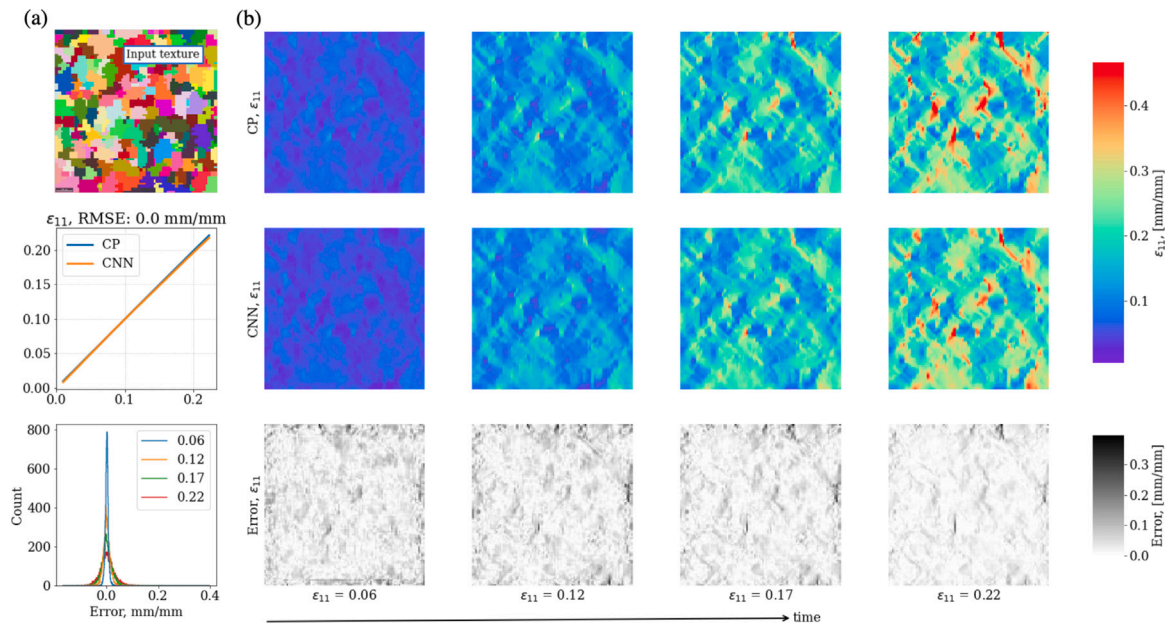


Fig. 13. Predictions for an AA5754 microstructure; (a): input microstructure, ϵ_{11} predictions for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh, and histogram showing the difference error distribution; (b): ϵ_{11} strain partitioning evolution for four strain levels. The first row displays crystal plasticity predictions, the second row displays convolutional neural network predictions, and the third row shows the error map, which displays the absolute element-wise difference error between the crystal plasticity and convolutional neural network predictions.

four different deformation levels, and (b) comparison of the strain evolution predicted by CPFEM and CNN for the same different deformation levels and the error map. It is important to notice that the microstructure of AA5754 is very different from the texture generated for the dataset. The typical examples of the texture in the dataset were presented in Figs. 10, 11, 12. Even though the morphology of the grains in the AA5754 microstructure is different from those found in the training set, the predictions of the CNN are in excellent agreement with crystal plasticity simulations. The CNN successfully predicted the strain hot spots, except for minor underpredictions of the strain intensity in the late stage of the deformation. The error map in the third row of Fig. 13, part (b), displays that the prediction errors are overall close to zero. The error histogram also demonstrates that most errors are concentrated around zero and are mostly less than 0.1 in their value for all deformation steps. Overall RMSE between CNN and CPFEM predictions for ϵ_{11} is equal to 0.02 (reported in Table 8) and the RMSE for the last time step is 0.03 (reported in Table 9) and is less than 5% of the maximum strain value in the mesh. The results for strains are in excellent agreement with CPFEM, which confirms the accuracy of the neural network model and its flexibility in predicting deformation for various heterogeneous microstructures.

Fig. 14 displays the evolution of σ_{11} stress during deformation process for the same initial microstructure. The second row of the figure's part (a) shows the average stress–strain curve. It was obtained by averaging the stress response across all elements in the mesh for each timestep. The RMSE error for average stress was equal to 3.18 MPa, which confirms the accuracy of overall predictions. For the element-wise comparison of predictions, the figure's first row of the part (b) shows the CPFEM predictions; then, the second row shows the CNN predictions, and the third row shows the element-wise absolute difference error for CPFEM and CNN predictions. The stress hot-spots are predicted for each timestep with excellent accuracy. The ADE heatmap displays that most of the errors are close to zeros, which confirms the accuracy of the CNN model. The histogram plot for the predictions (Fig. 14, (a), third row) also confirms that the CNN predictions are in excellent agreement with CPFEM simulations. Most of the difference errors are close to zero, and the great majority of the errors are less than 100 MPa in their absolute value. Overall RMSE between CNN and CPFEM predictions for σ_{11} is equal to 15.96 MPa (reported in Table 8) and the RMSE for the last time step is 24.75 MPa (reported in Table 9), which is less than 8% of the maximum stress value in the mesh. The results for strains are in excellent agreement with CPFEM, which confirms both the accuracy and flexibility of the proposed CNN model (see Fig. 15).

To fully verify the predictive capabilities of the convolutional neural network, another actual material microstructure was used for prediction. The microstructure under consideration is of the AA6061, and it was provided by Li et al. (2022). The pole figures of AA5754 and AA6061 microstructures are compared in Fig. 16, demonstrating the significant difference between textures presented in these two materials. The other substantial difference between the materials' microstructures is the grain morphology. The input microstructure of AA6061 can be observed in the first row Fig. 17, (a). The grains in the AA6061 exhibit overall shapes and sizes that are different from grains of the AA5754 microstructure (Fig. 13, (a), first row). The grains vary in size, unlike in artificially generated grains in the training data, and unlike in AA5754 microstructure, where grains are similar in their area. Some of the grains in the AA6061 microstructure exhibit different degrees of elongation, and grain morphology in this specimen is more heterogeneous than in AA5754. For this microstructure, ϵ_{11} strain and σ_{11} stress CPFEM and CNN predictions are compared in detail, and all other components are compared visually for the last timestep.

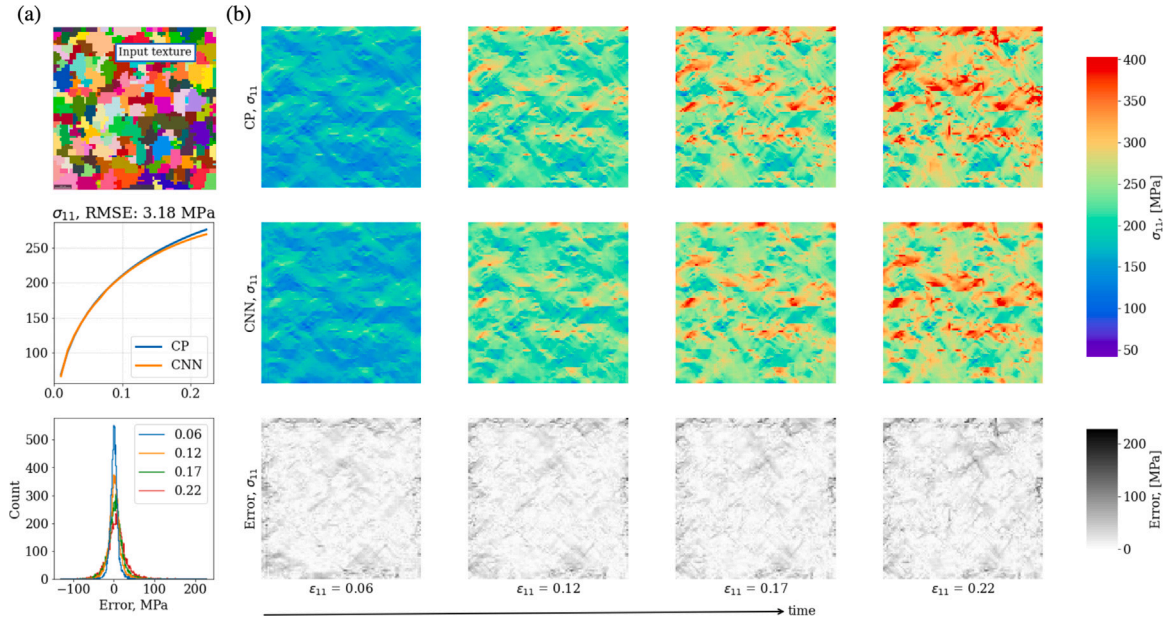


Fig. 14. Predictions for an AA5754 microstructure; (a): input microstructure, σ_{11} predictions for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh, and histogram showing the difference error distribution; (b): σ_{11} stress partitioning evolution for four strain levels. The first row displays crystal plasticity predictions, the second row displays convolutional neural network predictions, and the third row shows the error map, which displays the absolute element-wise difference error between the crystal plasticity and convolutional neural network predictions.

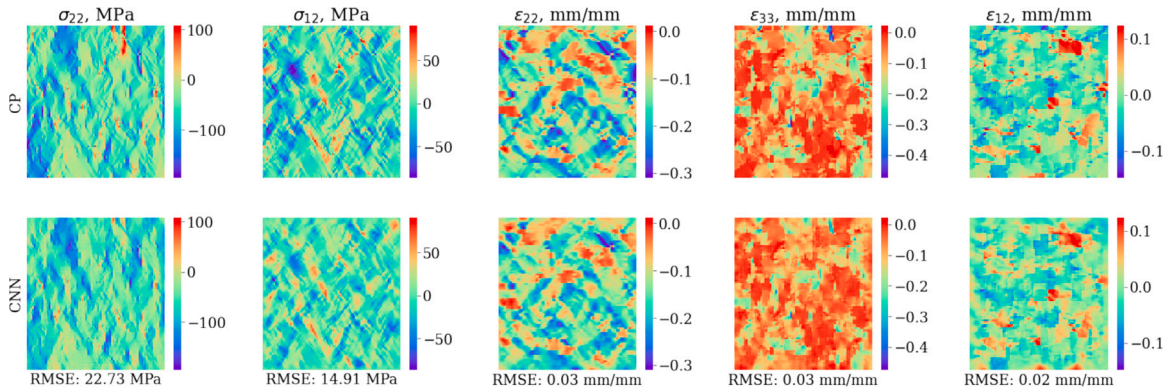


Fig. 15. Prediction for an AA5754 microstructure; the first row displays the crystal plasticity finite element predictions for the last timestep of the deformation, and the second row the convolutional neural network predictions for the last timestep of the deformation.

The average strain predictions ϵ_{11} for CPFEM and CNN are shown on Fig. 17, (a), second plot. The CNN model displays excellent predictive capabilities, and the error for the averaged predictions is negligible. The third plot in the (a) part of the figure shows the histogram for the difference error between predictions for four strain levels. The errors are mostly situated close to zero, and they are less than 0.1 in their absolute value. Strain partitioning comparison is shown on the (b) part of the figure, and the error map shows an absolute difference error. The red ellipse highlights the only hotspot that the machine learning model did not capture. The rest of the strain hotspots are captured with exceptional accuracy, which is confirmed by the error plot in the third row. The blue arrows highlight some of the successfully predicted strain hotspots within the partitioning. Overall RMSE for ϵ_{11} strain is equal to 0.02 (reported in Table 8) and the last timestep error for ϵ_{11} is equal to 0.03 (reported in Table 9), which confirm the consistency in the accuracies of the predictions. These results have demonstrated the excellent capability of the trained model to predict ϵ_{11} strain component mesh partitioning across the whole history of deformation.

The average stress predictions σ_{11} for CPFEM and CNN are shown on Fig. 18, (a), second plot. CNN displays excellent predictive capabilities, and the RMSE for the averaged flow curve is equal to 6.46 MPa. The third plot in the (a) part of the figure shows the histogram for the difference error between predictions for four strain levels. The errors are majorly situated close to zero in their absolute value. Stress partitioning comparison is shown in the (b) part of the figure, and the error map shows an absolute difference error. The blue ellipses highlight the major stress hotspots that evolved during the deformation, which were captured accurately

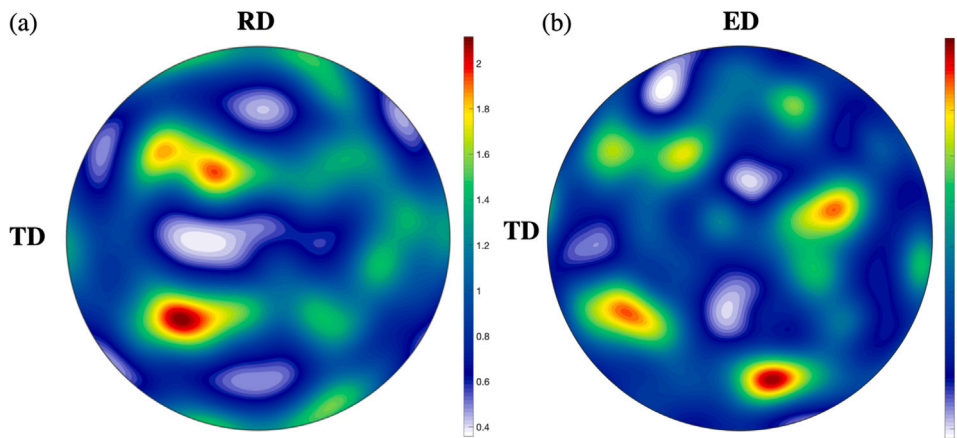


Fig. 16. (a) (111) pole figure for AA5754 microstructure and (b) (111) pole figure for AA6061 microstructure.

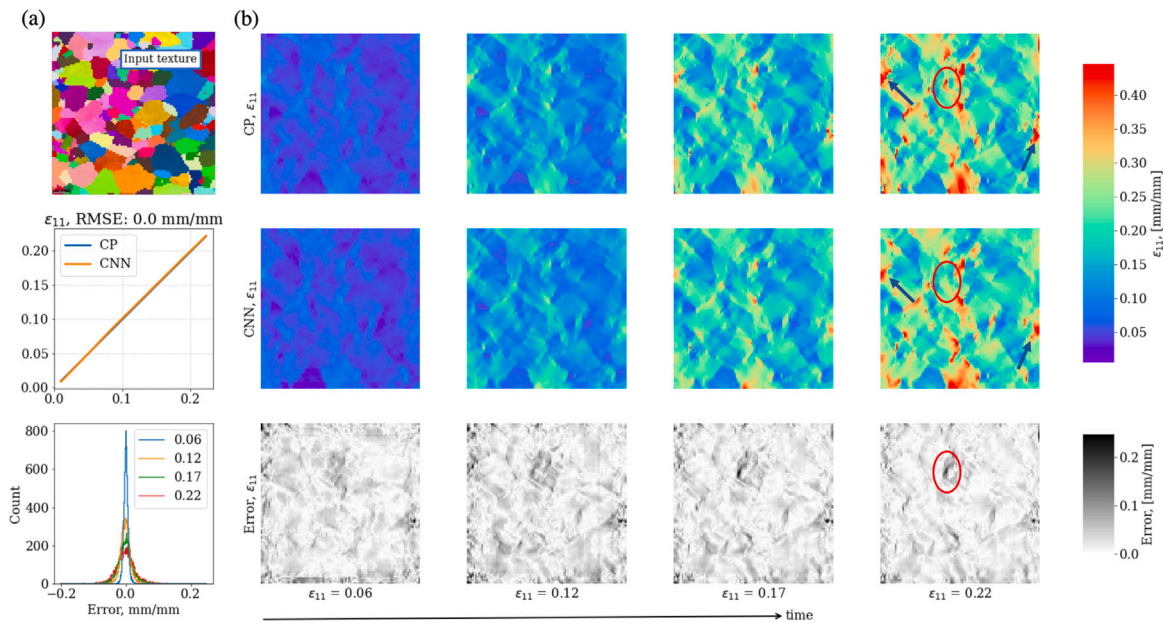


Fig. 17. Predictions for a AA6061 microstructure; (a): input microstructure, ϵ_{11} predictions for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh, and histogram showing the difference error distribution; (b): ϵ_{11} strain partitioning evolution for four strain levels. The first row displays crystal plasticity predictions, the second row displays convolutional neural network predictions, and the third row shows the error map, which displays the absolute element-wise difference error between the crystal plasticity and convolutional neural network predictions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

with the network. This accuracy is also confirmed by the absolute difference error plot in the third row. Overall RMSE for σ_{11} stress is equal to 17.49 MPa (reported in Table 8) and the last timestep error for σ_{11} is equal to 26.84 MPa (reported in Table 9), which confirm the consistency in the accuracies of the predictions. These results have demonstrated the excellent capability of the trained model to predict σ_{11} strain component mesh partitioning across the whole history of deformation for AA6061 alloy and validate the framework's capability to predict deformation behaviour for a wide range of materials.

The predictions for the other stress and strain tensor components were evaluated. As previously mentioned, only the last time step of the deformation process is visualised (Fig. 19). The overall accuracy of prediction is also clear from the visual representation. The predictions for the σ_{22} component have a slight underprediction across the mesh. Overall, RMSE for σ_{22} is equal to 18.46 MPa, and the RMSE for the last time step is 25.72 MPa. The prediction is visible σ_{12} very accurate, and all the stress hotspots are predicted correctly within the partitioning. Overall, RMSE for σ_{12} is equal to 9.86 MPa, and the RMSE for the last time step is 14.41 MPa. The predictions for the rest of the strain partitionings are performed exceptionally accurately. The overall RMSE for ϵ_{22} and ϵ_{33} is equal to 0.02, and overall RMSE for ϵ_{12} is as small as 0.01. For the last timestep, the RMSE for ϵ_{22} and ϵ_{33} is equal to 0.03, and the last timestep RMSE for ϵ_{12} is equal to 0.02. The overall RMSE for each of the variables is reported in the Table 8 and the last timestep

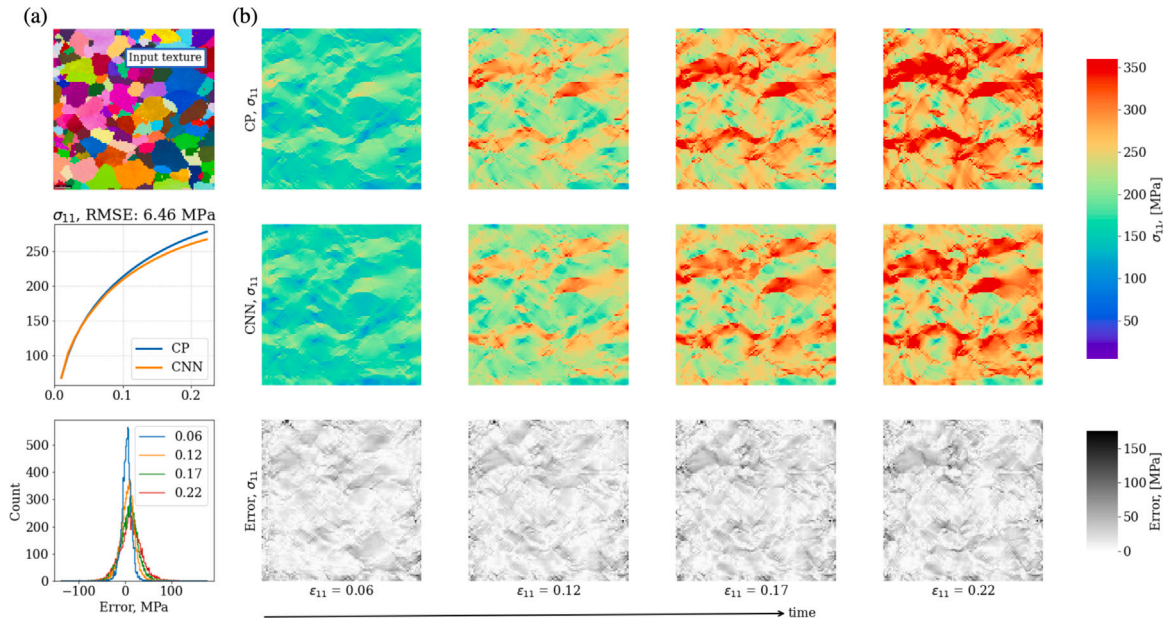


Fig. 18. Predictions for a AA6061 microstructure; (a): input microstructure, σ_{11} predictions for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh, and histogram showing the difference error distribution; (b): σ_{11} stress partitioning evolution for four strain levels. The first row displays crystal plasticity predictions, the second row displays convolutional neural network predictions, and the third row shows the error map, which displays the absolute element-wise difference error between the crystal plasticity and convolutional neural network predictions. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

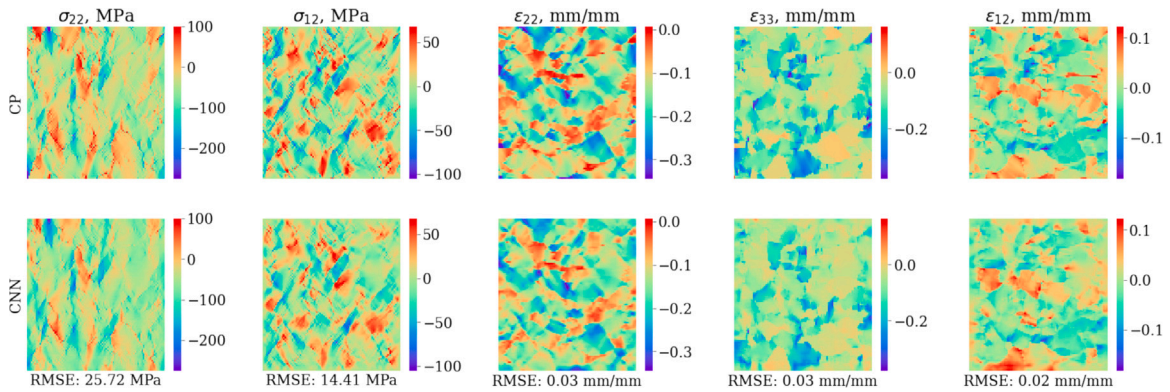


Fig. 19. Prediction for AA6061 microstructure; the first row displays the crystal plasticity finite element predictions for the last timestep of the deformation, and the second row the convolutional neural network predictions for the last timestep of the deformation.

Table 8

Application to AA5754 and AA6061: element-wise RMSE errors calculated across all timesteps for all predicted variables.

Case	σ_{11} , MPa	σ_{22} , MPa	σ_{12} , MPa	ϵ_{11}	ϵ_{22}	ϵ_{33}	ϵ_{12}
AA5754	15.96	15.78	10.06	0.02	0.02	0.02	0.01
AA6061	17.49	18.46	9.86	0.02	0.02	0.02	0.01

RMSE for each variable is reported in the Table 9. The visualisation of the results and the error values confirm the capability of the convolutional neural network to achieve accurate crystal plasticity predictions for real materials.

To conclude, the CNN was validated for AA5754 and AA6061 microstructures. Even though the network was never trained for the grain morphology in AA5754, the CNN predictions were in excellent agreement with the predictions of CPFEM simulations for this microstructure. The microstructure of AA6061 exhibited a completely new grain morphology and an overall texture. The CNN predictions were also accurate for AA6061, which demonstrated the flexibility of the proposed CNN for predictions for a wide range of materials.

Table 9

Application to AA5754 and AA6061: element-wise RMSE errors calculated only for the last timesteps for all predicted variables.

Case	σ_{11} , MPa	σ_{22} , MPa	σ_{12} , MPa	ϵ_{11}	ϵ_{22}	ϵ_{33}	ϵ_{12}
AA5754	24.75	22.73	14.91	0.03	0.03	0.03	0.02
AA6061	26.84	25.72	14.41	0.03	0.03	0.03	0.02

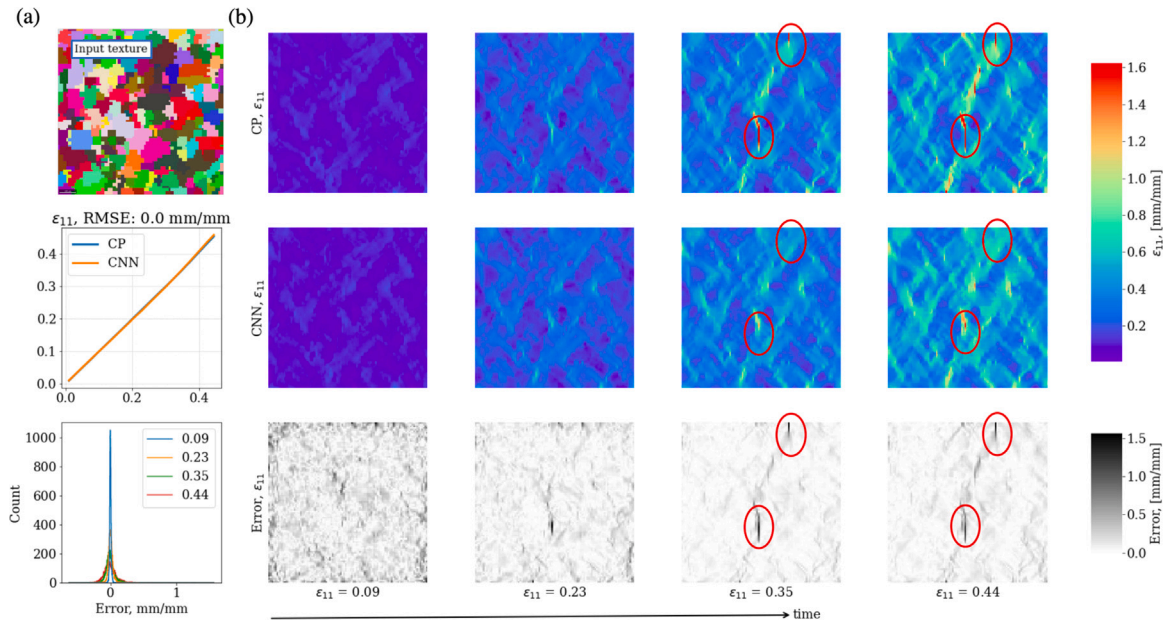


Fig. 20. Predictions for an AA5754 microstructure (extension to localisation predictions); (a): input microstructure, ϵ_{11} predictions for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh, and histogram showing the difference error distribution; (b): ϵ_{11} strain partitioning evolution for four strain levels. The first row displays crystal plasticity predictions, the second row displays convolutional neural network predictions, and the third row shows the error map, which displays the absolute element-wise difference error between the crystal plasticity and convolutional neural network predictions. Strain localisation intensity is underpredicted by the CNN, as highlighted by red ellipses. The predictions outside the localisation region and the averaged ϵ_{11} strain are captured accurately.

4.2. Extension to strain localisation prediction

In addition to the work presented above, the convolutional neural network's capability to predict strain localisation was investigated. The CPFEM data was simulated up to 44% of macroscopic ϵ_{11}^M strain, and the CNN was retrained on this data. Similarly to the previous training approach, the network was trained on 10,000 samples, validated on 1000 samples, and tested on 1000 samples. Final training and validation errors (normalised data) were equal to 1.3×10^{-3} after 60 training epochs, and the test error was equal to 1.2×10^{-3} . A low training error confirmed that the CNN learned the local strain distributions. However, CNN could not predict the correct intensity of the strain localisation formations. The average predictions were fairly accurate. For brevity and discussion purposes, only the example of predictions for AA5754 microstructure are presented. In detail, predictions for ϵ_{11} strain and σ_{11} stress components are analysed visually, and the rest components predictions were assessed with RMSE across all timesteps and for the last timestep of the deformation.

AA5754 microstructure was used to assess the predictive capabilities of the CNNs to predict strain localisation. Fig. 20, (a), shows the input microstructure, averaged strain response, and the difference errors for four strain levels in the form of a histogram. The averaged ϵ_{11} strain predicted with the neural network is predicted exceptionally accurately and has a minor error. The difference error plot also displays that most of the errors are situated in the proximity of zero, with the majority of the errors being less than 0.3 in their absolute value. However, the visualisations of the FE meshes (Fig. 20, (b)) reveal that the neural network, while capturing the localisation formation location, does not predict the correct intensity of hot spots, as highlighted by red ellipses on the plot. Outside of the localisation region, local ϵ_{11} strains are accurately captured by the convolutional neural network. The overall RMSE for ϵ_{11} component is equal to 0.16, and the last timestep RMSE is 0.26. Both of the errors are reported in the Table 10.

The analysis of the σ_{11} stress component prediction is demonstrated in Fig. 21. Part (a) of the figure displays the input microstructure, averaged stress error and the difference errors in the histogram. The averaged stress is captured well by the neural network, and the RMSE of the averaged predictions is 3.38 MPa. The CNN has successfully captured local stress relaxation, resulting in highly-accurate predictions of the averaged σ_{11} stress. The histogram confirms the CNN's accuracy by demonstrating that most local stress prediction errors are close to zero. The majority of the difference errors are less than 250 MPa in their absolute value.

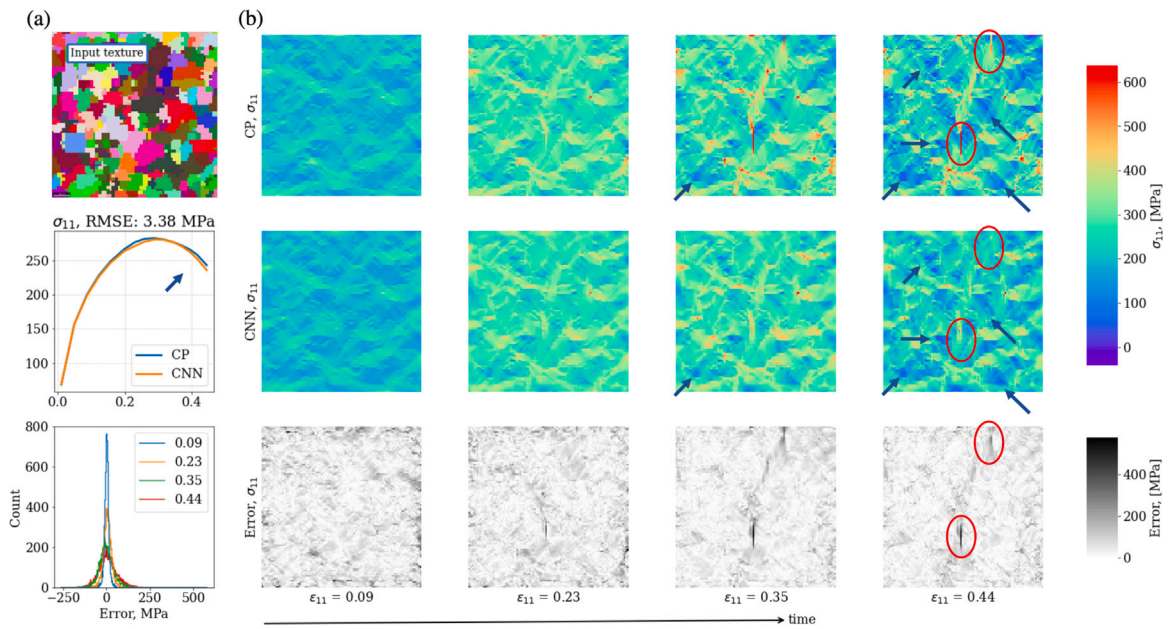


Fig. 21. Predictions for an AA5754 microstructure (extension to localisation predictions); (a): input microstructure, σ_{11} predictions for crystal plasticity and convolutional neural network predictions averaged across all elements of the mesh, and histogram showing the difference error distribution; (b): σ_{11} strain partitioning evolution for four strain levels. The first row displays crystal plasticity predictions, the second row displays convolutional neural network predictions, and the third row shows the error map, which displays the absolute element-wise difference error between the crystal plasticity and convolutional neural network predictions. The intensity of stress localisation is underpredicted, as highlighted with red ellipses. The CNN accurately captures local stress relaxation and the averaged stress response, as demonstrated by the arrows. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

Table 10

Element-wise RMSE errors calculated across all timesteps for all predicted variables (extension to localisation predictions).

Case	σ_{11} , MPa	σ_{22} , MPa	σ_{12} , MPa	ϵ_{11}	ϵ_{22}	ϵ_{33}	ϵ_{12}
All timesteps	80.8	60.36	37.51	0.16	0.12	0.11	0.07
Last timestep	116.39	77.71	53.77	0.26	0.19	0.16	0.12

As in the case with the prediction of the ϵ_{11} strain component, the predictions of the stress hotspot location are correct, but the intensity of the predicted value is not captured accurately, as highlighted with red ellipses on the figure. The other areas of the FE mesh are in excellent agreement with crystal plasticity simulations. Notably, the CNN predicted the local stress relaxation quite well, as highlighted with blue arrows on the figure. The accuracy of prediction is also reflected in the averaged flow curve: the stress decreases after the material's maximum resistance to loading is achieved. The error map validates the observations: most errors are close to zero, apart from the localisation spots. The overall RMSE for the σ_{11} predictions was equal to 80.8 MPa, and the last timestep RMSE was equal to 166.39 MPa. The errors are reported in the Table 10.

The errors for the other variables are reported in the Table 10. Similarly to the results for ϵ_{11} and σ_{11} , the RMSEs for σ_{22} , σ_{12} , ϵ_{22} , ϵ_{33} , and ϵ_{12} are higher than in the cases where deformation is predicted up to 22%. The authors attempted to train deeper networks, adding more data to the training set, which did not help achieve better localisation predictions. The other types of networks that can be considered for application of the localisation predictions could be convolutional long short-term memory networks (CNN-LSTM), a recurrent neural network. They could be beneficial for predictions of spatial-sequential data such as deformation evolution of finite element mesh. These networks were not considered in this research as they are out of the scope of this work.

4.3. Runtime comparison & model limitations

4.3.1. Runtime comparison

The results have demonstrated the predictive capabilities of the proposed CNN to predict stress and strain partitioning of FE mesh. Besides the predictive accuracy, CNNs have another beneficial quality: computational efficiency. Crystal plasticity simulations are computationally demanding (Brahme et al., 2011b), especially when it comes to finite element applications. Crystal plasticity finite element models' computational time increases exponentially with the increase of elements in the FE mesh (Ali et al., 2019). The exponential increase in computational time would not be the case for neural networks due to the relative simplicity of the performed computations in the networks. CNN computations include operations such as matrix multiplication, addition, and

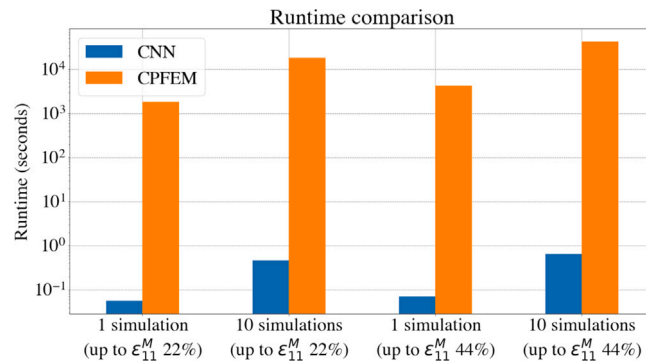


Fig. 22. Runtime comparison between CPFEM and CNN. Vertical axis is displayed using logarithmic scale to emphasise the time difference between the models.

Table 11

Summary of runtime for different processes used in this research.

Process	Number of samples	Time
Data generation (CPFEM)	12,000 samples, up to $\epsilon_{11}^M = 22\%$	6000 CPU hours
CNN training time	100 epochs with 10,000 training and 1000 validation samples	37 GPU hours
Forecasting with CPFEM	1 sample, up to $\epsilon_{11}^M = 22\%$	30 CPU minutes
Forecasting with CNN	1 sample, up to $\epsilon_{11}^M = 22\%$	56 GPU milliseconds

application of non-linearity to the weights in a layer, while crystal plasticity methods require solving highly non-linear computational equations. Therefore the usability of the networks becomes evident for such problems as material design, as multiple microstructures' deformation behaviour can be evaluated by neural networks simultaneously in seconds or minutes. Crystal plasticity simulations would require computationally demanding simulations, usually performed on high-performance computing clusters. If the number of microstructures requiring deformation behaviour evaluation is large, crystal plasticity simulation may not be feasible due to computational limitations, while machine learning methods are fast and becoming more and more accessible nowadays.

A number of studies compared the runtime of crystal plasticity compared to machine learning models (Mangal and Holm, 2018b, 2019; Ali et al., 2019; Acar, 2020; Ibragimova et al., 2021) and highlighted the computational time superiority of machine learning models to crystal plasticity models. The computational time comparison has also been conducted in this study and confirmed the computational efficiency of the neural networks compared to crystal plasticity methods. Fig. 22 demonstrated runtime comparison for CPFEM and CNN models. The comparison in seconds for 1 and 10 simulations is demonstrated for predictions up to 22% and 44% of ϵ_{11}^M strain. A logarithmic scale was used to highlight the magnitudes of difference in the runtime. For example, to predict local stress and strain values up to 22% strain, the CPFEM model takes 30 min for one simulation and 300 min for 10 simulations, while the CNN model takes 56.3 ms and 470 ms correspondingly. For 1 and 10 simulations for localisation predictions, the CPFEM model requires 71 minute and 710 min, while CNN requires 71 ms and 650 ms correspondingly.

Table 11 presents a runtime summary of the processes presented in the current research. Training data generation was the most time-consuming process and resulted in utilising 6000 central processing unit hours (CPU hours) to collect 12,000 samples. After the dataset was collected, it was used for training. The training time of the selected CNN (Fig. 8, Section 3.2) was approximately 37 h on graphics processing unit (GPU). The training dataset was comprised of 10,000 samples, and after each of 100 training epochs, a validation on 1000 samples was performed. To forecast one simulation up to $\epsilon_{11}^M = 22\%$, CPFEM required 30 CPU-minutes, and CNN required 56 GPU-milliseconds. CPFEM time was calculated in CPU hours because the CPFEM algorithm was optimised for use on CPU, and CNN time was estimated in GPU hours as training and forecast CNN algorithms were optimised for use on GPUs. Therefore, machine learning enables enormous time savings and results in over 99.9% faster predictions.

4.3.2. Limitations & suggestions for future work

While the approach presented in this work provides a significant speed-up for crystal plasticity-based calculations, it still has the following limitations (in terms of accuracy) in: (i) capturing the overall stress softening, (ii) the magnitude of localisation, and (iii) will simulate the strain path $\rho = \frac{\epsilon_{22}}{\epsilon_{11}} = -0.5$.

The limitation of the model in predicting the overall stress softening is due to the training dataset not including enough examples. The extension of the model presented in Section 4.2 provides a solution by including more training examples with localisation and the retrained model does capture the stress softening. Similarly, the model is trained for $\rho = \frac{\epsilon_{22}}{\epsilon_{11}} = -0.5$ and thus it only predicts the response for this strain path. The model can be extended to predict material response to various strain paths, by introducing another input feature, ρ (ratio of the minor strain to the major strain).

As mentioned above, the model does not capture the exact intensity of localisation. To address this limitation, a different machine learning algorithm can be used, such as a recurrent neural networks, which can learn the time dependent behaviour (Kohar et al., 2021).

The current model is trained on input microstructure with fixed dimensions, viz. 100×100 element mesh. Hence, the model can only predict the response of microstructures with this input dimension. To extend the model to accept any input dimension a sliding-window approach can be implemented to predict local values of stresses and strain for a given region of a material microstructure. A similar approach was initially presented for a bio-medical application in Ciresan et al. (2012).

The proposed model in this research can be extended for three-dimensional applications with the use of three-dimensional CNNs. The model can be extended to predict the material response for various strain paths. This could be achieved by introducing another input feature representing a strain path (loading condition). The other input features, such as new physical parameters, can also be introduced into the model. Transfer learning methods (e.g. using pre-trained weights of the neural network as initial new weights as a whole or partially) can be used to train new networks to accelerate the training procedure.

5. Summary & conclusions

In this paper, a CNN-based framework to achieve high-fidelity predictions of local stress and strain evolution is proposed. The proposed framework resulted in training an optimal CNN on a CPFEM dataset and included: feature engineering, synthesis of microstructures, data preprocessing, dataset design, and optimal architecture search. The important summary points, observations, and conclusions are as follows:

- The result of the framework was a deep convolutional neural network which enabled accurate crystal plasticity finite element predictions for materials.
- The training dataset consisted of synthetically generated microstructures and two material parameters: initial hardness and initial hardening modulus. The microstructures were generated using Voronoi tessellations; each microstructure was unique and contained a varying number of grains. The material hardening parameters were sampled using Sobol sequences. The orientations for generated microstructures were randomly sampled from the dataset of orientations. The orientation dataset was generated using AA5754 EBSD scan and contained 8750 orientations.
- The CNN architecture was selected based on the ten different architecture trial training on the partial dataset (5000 samples) for 15 epochs. Then, the best performing neural network was trained on the complete dataset for 100 training epochs. The training consisted of 10,000 samples, and the validation and test sets consisted of 1000 samples each. The training dataset consisted of CPFEM simulation data up to $\epsilon_{11}^M = 22\%$ of strain.
- The CNN's predictive capabilities were validated using a test set containing completely new synthetically generated microstructures. Then the CNN was validated for two actual material microstructures (AA5754 and AA6061). The grain morphology within the AA5754 microstructure was different from the typical synthetically generated microstructure. The texture and the grain morphology of the AA6061 microstructure were also completely new to the CNN.
- The network has demonstrated excellent agreement with crystal plasticity simulations for test sets and both real material samples.
- Test set median prediction error (RMSE across all timesteps) for σ_{11} was 16.14 MPa and for ϵ_{11} was 0.02. Test set worst prediction error (RMSE across all timesteps) for σ_{11} was 22.69 MPa and for ϵ_{11} was 0.04. Test set best prediction error (RMSE across all timesteps) for σ_{11} was 15.37 MPa and for ϵ_{11} was 0.02. The other components' errors were also reported. In the worst prediction case, the CNN could not predict the localisation which appeared earlier during the deformation process due to specific texture morphology. The cases with localisation formations were not common within the dataset. Therefore the network did not learn the localisation behaviour, resulting in higher error. The median and best errors were close in values, which confirms the accuracy of the neural network. Visually, the meshes also confirm the excellent agreement between CPFEM and CNN predictions.
- The network demonstrated excellent predictive capabilities for real AA5754 microstructure. The orientation distribution was similar to the orientation distributions in microstructures from the synthesised dataset, but the grain morphology was different. The real AA5754 grain morphology was rather distorted than a smooth and similar size and shape of grains in the synthetically generated material. Despite that microstructural difference, the network predictions demonstrated excellent agreement with crystal plasticity results. The predictions were compared visually for four strain levels, and the agreement between the results was evident. The prediction error (RMSE across all timesteps) for σ_{11} was 15.96 MPa and for ϵ_{11} was 0.02. The other components for strain and strain were also reported. The error results were comparable to the best test case predictions, thus confirming the accuracy of the neural network.
- The network also demonstrated excellent predictive capabilities for real AA6061 microstructure. The orientation distribution was different for this microstructure compared to the orientation distributions present in the synthesised dataset. The grain morphology differed from both the synthesised dataset and AA5754 microstructure. Despite that microstructural difference, the network predictions demonstrated excellent agreement with crystal plasticity results. The predictions were compared visually for four strain levels, and the agreement between the results was evident. The prediction error (RMSE across all timesteps) for σ_{11} was 17.49 MPa and for ϵ_{11} was 0.02. The other components for strain and strain were also reported. The error results were comparable to the best test case predictions, thus confirming the accuracy of the neural network and its applicability outside of the training dataset.
- To further extend the ability of the proposed CNN to predict localisation, the network was re-trained on the CPFEM simulations for data up to $\epsilon_{11}^M = 44\%$ of strain. The proposed CNN successfully captured average stress and strain responses. Overall, the predictive performance of the CNN showed good agreement with CP simulations, except for the exact magnitude of the localisation intensity, including unloading behaviour (stress softening) predictions.

- The presented approach to CPFEM predictions has been computationally efficient compared to the crystal plasticity finite element model. The trained convolutional neural network has shown up to 99.9% computational speedup: up to 65 ms versus up to 11 h and 50 min for ten simulations. As a direction to further study, such machine learning models can be applied to the microstructural design problems and save a considerable amount of time on crystal plasticity simulations, as conventional crystal plasticity models are very computationally demanding.

CRedit authorship contribution statement

Olga Ibragimova: Conceptualisation, Investigation, Methodology, Formal analysis, Software, Visualisation, Writing – original draft. **Abhijit Brahme:** Conceptualisation, Methodology, Supervision, Writing - review & editing. **Waqas Muhammad:** Conceptualisation, Investigation, Methodology, Supervision, Writing - review & editing. **Daniel Connolly:** Conceptualisation, Methodology. **Julie Lévesque:** Supervision, Project administration, Funding acquisition. **Kaan Inal:** Supervision, Project administration, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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